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Customer concentration and the readability of 10-K reports

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ABSTRACT

We investigate the relationship between customer concentration and the readability of suppliers' financial disclosures. Using a large sample of 9554 US-listed firms from 1994 to 2020, we find that suppliers with a more concentrated customer base produce lower quality 10-K reports. This negative association is driven by managerial obfuscation motives and is more pronounced for suppliers that pay excess CEO compensation, face lower customer switching costs, make higher relationship-specific investments and have greater board independence. Suppliers with high customer concentration produce better quality reports when they face increased institutional monitoring. We mitigate potential endogeneity issues using an instrumental variable, propensity score, entropy balance matching, and Oster's (2019) test for omitted variable bias. Our results are robust to several alternative proxies for both customer concentration and 10-K report readability. Our findings provide robust empirical evidence that suppliers with a highly concentrated customer base produce less readable financial information.

1. Introduction

Academics and practitioners are paying increasing attention to the impact of customer-supplier relationships on firm operations. Customer concentration, described as the degree of a supplier firm's financial dependence on its major customers, is considered a crucial aspect in this relationship because of the strength of a supplier's dependence on a concentrated or diversified customer base and its impact on global value chains (Huang et al., 2016; Kim & Zhu, 2018; Tangpong et al., 2008; Zhu et al., 2021). Previous research demonstrates that supplier accounting practices, such as earnings management (Raman & Shahrur, 2008), disclosure of litigation losses (Cen et al., 2017) and tax avoidance (Cao et al., 2020; Huang et al., 2016) are significantly affected by customer concentration.

In this paper, we examine the effect of customer concentration on the readability of suppliers' 10-K reports because they are a critical aspect of financial reporting quality. Research on the readability of 10-K reports has received substantial consideration in the literature and in regulatory proposals (Cassell et al., 2019; Li, 2008, 2019; Lo et al., 2017). The language clarity in 10-K reports is crucial for stakeholders' decision-making, particularly for regular investors with limited information processing ability to grasp an accurate understanding of a company's past performance, current status, and future prospects (Clarkson et al., 1994; Lawrence, 2013).

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Because language and text choices have such a substantial impact on stakeholder decision-making, producing understandable, transparent financial statements demonstrates an organization's obligation to fulfil ethical principles (Ben-Amar & Belgacem, 2018). Customers care about the readability of their suppliers' 10-K reports because they are important sources of forward-looking information about the financial viability of a company. For instance, companies dependent on suppliers offering unique products will face switching costs if a supplier is unable to continue operations into the future (Bowen et al., 1995). Additionally, the value of relationship-specific investments made by customers in suppliers is tied to the supplier firm's future prospects (Cornell & Shapiro, 1987). Prior research suggests that customers rely on suppliers' financial information when negotiating contracts (Crawford et al., 2020).[1] Anecdotal evidence suggests that environmentally responsible customers are interested in knowing supplier firms' initiatives on decarbonization and deforestation to ensure sustainability (Mace, 2021).[2] The readability of suppliers' 10-K reports, in terms of language complexity, is therefore an important consideration in the customer-supplier relationship.

We hypothesize that customer concentration could influence the readability of 10-K reports both positively and negatively. On the one hand, suppliers facing higher customer concentration may produce lower quality financial reports to hold existing customers by withholding bad news and obfuscating disclosures. Loss of a key customer for such suppliers could lead to a sharp drop in the supplier's cashflow (Dhaliwal et al., 2016; Huang et al., 2016). In addition, such firms often undertake sizable relationship-specific investments to particularly cater specifically to the needs of major customers. Such investments could result in significant losses if the supplier loses the major customer, greatly increasing the supplier's risk of financial distress (Campello & Gao, 2017; Dong et al., 2021; Kale & Shahrur, 2007; Titman & Wessels, 1988). Given the business and cash flow concerns that they face, previous research indicates that suppliers with concentrated customers are more inclined to hoard bad news (Lee et al., 2020; Ma et al., 2020). Furthermore, though prior research often interprets linguistic complexity in firms' disclosures as a sign of managerial obfuscation, Bushee et al. (2018) reveal that linguistic complexity encompasses both obfuscation and information factors that affect information asymmetry in opposite directions.[3] In this regard, even if we argue that a dependent supplier does not intentionally create an asymmetric information environment for obfuscatory purposes, the higher information content of linguistic complexity resulting from overly technical disclosures (also referred to as information overload) could also lead to less readable statements (Campbell et al., 2014; Lee, 2012). For example, major customers may use their negotiating power to influence suppliers to disclose any financial difficulties and uncertainties they are facing (Bloomfield, 2008; Campbell et al., 2014; Lee, 2012). Such information content may lead to over-technical disclosures and less readable 10-K reports (Bushee et al., 2018; Bushee & Huang, 2024; Skinner, 2024). Based on these arguments, we expect suppliers with high customer concentration to produce less readable financial reports.

Conversely, a supplier firm's emphasis on serving a concentrated customer base can strengthen the relationship between the parties, potentially enhancing its financial performance and reducing cash flow and business risks for the supplier (Irvine et al., 2016; Patatoukas, 2012). Greater monitoring by major customers is likely to improve a supplier's corporate governance practices, reduce disclosure obfuscation and meet the major customer's environmental demands (Chakrabarty et al., 2018; Johnsen & Lacoste, 2016; Zheng et al., 2023; Zhu et al., 2021). According to the certification hypothesis, dependent suppliers are likely to be vigilantly monitored and certified by concentrated customers to reduce information asymmetry (Cai & Zhu, 2020; Cen et al., 2016). The customer-supplier relationship may also induce suppliers to have conservative reporting practices and higher audit quality (Hui et al., 2012; Krishnan et al., 2019). Through these incentives, reduced information asymmetry for suppliers with a concentrated customer base would likely improve the quality of their financial reports through better readability. These conflicting theoretical arguments suggest that the impact of customer concentration on the readability of 10-K reports is the empirical question that we investigate in this paper.

Using a substantial sample of 9554 US-listed public firms from 1994 to 2020, we investigate the relationship between suppliers' customer concentration and the readability of the 10-K reports. Our primary indicators of customer concentration are the proportion of a supplier's total sales to major customers (Major Customer Sales) and the Herfindahl-Hirschman Index (Major Customer HHI) to gauge the concentration of sales. Additionally, we use a dummy variable (Major Customer) set to one if a supplier reports at least one corporate customer responsible for 10 % or more of its annual revenue, and zero otherwise (Dhaliwal et al., 2016; Huang et al., 2016). To assess annual report readability, we use the BOG Index and the natural logarithm of the 10-K report file size as key dependent variables (Bonsall IV et al., 2017; Hasan, 2020). Our findings indicate a positive, significant relationship between customer concentration and the BOG index. Since a higher BOG index value and larger file size imply reduced readability, this suggests that suppliers with concentrated customers tend to have reduced readability 10-K reports. Our findings are also economically significant. For example, a 1 % increase in major customer HHI leads to 1.11 % and 19.19 % increase in the Bog Index and file size of 10-K reports for average firms, respectively. We confirm the robustness of our results by applying an alternative readability measure (the Fog Index) and obtain consistent findings.

Our baseline findings could be biased due to endogeneity concerns arising from omitted variables and model misspecification issues. We address these endogeneity concerns by conducting a Two-Stage Least Squares (2SLS) regression using an adverse customer industry shock as an instrumental variable (IV). We consider this a credible instrument because industry shocks affecting customers are

[1] See Crawford et al. (2020) Appendix 1 for excerpts of these contracts.

[2] Tesco, one of the largest UK-based retail businesses, recently implemented a supply chain programme that links to the sustainability performance of their suppliers in providing financial support. Tesco will provide lower financing rates to suppliers addressing environmental impacts in their disclosures. Thus, suppliers dependent on Tesco will benefit from better readability of their annual reports, highlighting the significance of 10-K readability.

[3] We thank the anonymous reviewers for this valuable suggestion.

likely exogenous to supplier firms and are less likely to impact them through channels other than supply chain relationship, thus meeting the exclusion criterion for a valid instrument. Such negative industry shocks can reduce customer demand and create potential instability through customer exits or mergers and acquisitions, satisfying the relevance criteria. Thus, suppliers with a concentrated customer base are expected to be significantly negatively affected by adverse customer industry shocks (Albuquerque et al., 2014; Bhattacharyya & Nain, 2011; Ma et al., 2020). We also conduct Oster's (2019) test for omitted variable bias and propensity score matching (PSM) and entropy balancing using matched samples to tackle model misspecification issues and sample selection bias. The 2SLS, Oster (2019) test, the PSM and entropy results, all validate the consistently negative linkage between customer concentration and the readability of financial reports.

Based on Bushee et al. (2018), we examine the motive for reduced readability of suppliers' 10-K reports. We first examine the role of managerial linguistic complexity. There are two latent components of linguistic complexity: obfuscation and the information content of disclosures. The obfuscation component acts as a proxy for the use of unnecessarily complex words to reduce the understandability of disclosures. The information component refers to the quantity of informative technical disclosures provided by managers during conference calls (Bushee et al., 2018; Bushee & Huang, 2024; Gu et al., 2024; Hooghiemstra et al., 2017). We find that concentrated supplier firms reduce the readability of their 10-K reports for the obfuscation motive rather than a higher volume of technical information from the managers' linguistic complexity. We also examine the mediating role of supplier cash flow volatility in the relationship. Using a structural equation model (SEM) approach, we find evidence that suppliers with concentrated customers are motivated to reduce the readability of 10-K reports to manage potential the cash flow volatility from losing customers, consistent with previous studies (Dhaliwal et al., 2016; Huang et al., 2016).

We also explore cross-sectional variation in the association between customer concentration and the readability of 10-K reports by considering related factors such as CEO excess compensation, institutional ownership, customer switching costs, board independence and relationship-specific investment (Dhaliwal et al., 2016; Jain et al., 2023; Kale & Shahrur, 2007; Rahman & Kabir, 2023). We document that the negative link between customer concentration and 10-K report readability is more evident when the supplier firm CEO receives excess compensation (motivating opportunistic behaviour to pursue private benefit), has lower switching costs (motivating suppliers to obfuscate and delay bad news from reaching their major customers), is controlled by an independent board (promoting a greater incentive for managers to obfuscate when dealing with the board) and makes more relationship-specific investments (motivated by the need to retain major customers). However, we find the opposite relationship for firms with greater institutional ownership, driven by managers having limited opportunities to obfuscate disclosures because of institutional monitoring. These results support the prediction that such channels have a greater impact on the readability of 10-K reports. Finally, we examine the firm-level outcomes of greater customer concentration and poor 10-K report readability. We find that, in such a scenario, suppliers face lower firm valuation and greater stock price crash risk, highlighting the importance for suppliers to diversify their customer base and improve the readability of their disclosures.

The study contributes to the literature in several ways. First, it provides new evidence in the 10-K readability literature and contributes to existing knowledge by highlighting how business risks associated with suppliers' dependence on concentrated customers affect their financial reporting quality. Previous studies examining the determinants of the readability of firm's 10-K reports consider mostly internal factors such as firm earnings (Li, 2008; Lo et al., 2017), managerial ability (Hasan, 2020), board gender diversity (Ginesti et al., 2018; Hasan, 2020) and business strategy (Habib & Hasan, 2020; Lim et al., 2018). By examining the impact of customer concentration on the readability of 10-K reports, an important aspect of the supply chain process, we demonstrate that readability is significantly affected by risks in the supply chain relationships. Prior studies show that customer concentration leads managers to withhold bad news (Chen et al., 2024), but we go beyond that association by examining what motivates managers to reduce financial report quality. Most importantly, we examine the motives for producing poor quality disclosures and find that they are driven by managerial obfuscation rather than information content (Bushee et al., 2018). As managerial disclosures can significantly affect information asymmetry with stakeholders, our study highlights the managerial motivation for producing less readable 10-K reports when supplier firms have high customer concentration.

Second, our study adds to the ongoing discussion regarding the potential effects of significant customer concentration on a firm's financial reporting practices. Though prior studies explore its impact on earnings management (Raman & Shahrur, 2008), tax avoidance (Huang et al., 2016), and disclosure of litigation losses (Cen et al., 2017), our results indicate that suppliers' dependence on concentrated customers decreases the readability of their 10-K reports, which is a crucial element of financial reporting quality. Our study offers detailed evidence by highlighting the importance of various context factors—such as excess CEO compensation, customer switching costs, board independence, institutional ownership, and relationship-specific investments—in understanding the link between customer concentration and the readability of firm disclosures (Cen et al., 2017; Chakrabarty et al., 2018; Dong et al., 2021; Ertugrul et al., 2017; Johnsen & Lacoste, 2016; Kale & Shahrur, 2007). We demonstrate that these mechanisms play a critical role in shaping how suppliers disclose information. Our findings enhance the understanding of how these context elements affect the relationship between customer concentration and the readability of firm disclosures.

Third, our study provides evidence of the consequences of the reduced 10-K report readability for supplier firm performance outcomes. Reinforcing the negative impact of customer concentration on supplier firms' financial reporting quality, we show that such factors are significantly detrimental to suppliers' financial performance (Hasan, 2020; Ma et al., 2020). Their negative impact on firm valuation and stock price crash risk significantly increases the importance of financial reporting and disclosure to stakeholders in assessing how firms and their managers structure their operations and drive firm value.

The remainder of the paper is structured as follows. The literature review and hypothesis formulation are presented in section 2 and the data and research design are discussed in section 3. We present the empirical results and endogeneity tests in section 4, and the motivating factors driving the relationship in section 5. The study concludes in Section 6.

2. Literature review and hypotheses

2.1. Annual report readability

Annual reports (known as 10-K reports in the United States) submitted by publicly traded companies are important sources of information for investors and stakeholders. These narrative disclosures rely heavily on qualitative information, which accounts for approximately 80 % of the 10-K reports (Lo et al., 2017). The readability of 10-K reports in effectively communicating financial statement information is a major concern of stakeholders because they narrate the company's future prospects and risk exposure and bridge the gap between company insiders and external stakeholders (Clarkson et al., 1994; Courtis, 1986; Loughran & McDonald, 2014). Despite their importance, these narrative disclosures are often marred by complexity and linguistic opacity that are frequently the result of managerial practices (Bonsall & Miller, 2017; Hasan, 2020). This can hinder the effective exchange of information between insiders and the external stakeholders who rely on these reports to make informed decisions.

Several studies have explored the relationship between various firm-specific factors and their impact on the readability of annual reports. For example, a company's financial performance has a significant impact on how it presents information in its 10-K reports (Li, 2008; Lo et al., 2017). The competence of a firm's management team can also affect the clarity and comprehensibility of the reports (Hasan, 2020). Board gender diversity (Ginesti et al., 2018; Hasan, 2020), business strategy (Habib & Hasan, 2020; Lim et al., 2018), political corruption (Xu et al., 2022), employment quality (Tan et al., 2023) and tax avoidance (Nguyen, 2021) are some of the other important internal factors that affect the readability of 10-K reports. In addition to identifying these determinants, previous studies have also examined the effects of 10-K report readability on diverse aspects of corporate performance and financial markets. Readability significantly influences firms' investment efficiency (Biddle et al., 2009), analysts' following (Lehavy, Li, & Merkley, 2011), loan contracts (Ertugrul et al., 2017), bond ratings (Bonsall & Miller, 2017) and agency costs – emphasizing its role in reducing information asymmetry between managers and stakeholders (Luo, Li, & Chen, 2018).

2.2. Customer concentration

Early economic research on the dynamics between suppliers and major customers emphasized that significant customers often exert pressure on their dependent suppliers, leading to concessions such as lower prices, extended trade credit terms, and the retention of excess inventory. Conversely, suppliers with a concentrated customer base can realize operational efficiencies through collaboration on marketing and advertising initiatives, enhancing the effectiveness of their selling expenses (Chen & Roma, 2011; Kalwani & Narayandas, 1995; Oliveira et al., 2017; Porter, 1974). Notably, around 45 % of publicly traded companies acknowledge their reliance on one or more major customers for a substantial portion of their revenue (Ellis et al., 2012). A key strategy for exercising power involves keeping the supplier unaware of the intentions of major customers (Galbraith et al., 1994). This uncertainty stems from the fact that major customers often design their orders in ways that dictate the production and investment strategies of suppliers with a concentrated customer base. This increased bargaining power enables major customers to obtain sensitive information from suppliers at a lower cost and secure financial advantages such as favourable trade credit terms (Crawford et al., 2020; Wali Ullah et al., 2024).

Accounting practices such as profitability (Hui et al., 2019), earnings management (Raman & Shahrur, 2008), litigation loss disclosures (Cen et al., 2017), mergers and acquisitions (Dong et al., 2021) and tax avoidance (Cao et al., 2020; Huang et al., 2016), are all significantly affected by the customer concentration level. Research on industrial organizations indicates that higher industry-level customer concentration is typically linked to lower profits for suppliers, although a recent study has provided contradictory findings (Bhattacharyya & Nain, 2011). Specifically, supplier firms with a concentrated customer base tend to enjoy higher profit margins and asset turnover (Patatoukas, 2012). Some researchers argue that customer concentration encourages R&D investment and increases innovation intensity (Krolikowski & Yuan, 2017), aids in the success of new products in the market (Gruner & Homburg, 2000), and enhances inventory management efficiency (Ak & Patatoukas, 2016). Conversely, the "collaboration relationship life-cycle" hypothesis posits that the relationship between major customer concentration and firm profitability is initially negative when the partnership is new but becomes positive as the relationship develops over time (Irvine et al., 2016). Additionally, customer concentration has been associated with increased costs of equity capital (Dhaliwal et al., 2016) and cost of debt (Campello & Gao, 2017), a greater likelihood of future stock price crash risk (Lee et al., 2020; Ma et al., 2020), and poor performance in mergers and acquisitions (Dong et al., 2021). However, the relationship between customer base concentration and the readability of annual reports remains an empirical issue, with no clear ex-ante conclusion.

2.3. Hypotheses development

As key stakeholders, corporate customers have a significant influence on how their suppliers conduct their business. The Statement of Financial Accounting Standards (SFAS) No. 131 (formerly SFAS No. 14) mandates that supplier firms disclose information regarding their major customers because these customers represent "a significant concentration of risk" (Dhaliwal et al., 2016). Firms face not only regulatory pressure to provide details of their major customers but also concerns about the risks associated with these customers.[4]

[4] In 2011, Tenneco Inc. disclosed in its annual report that "the loss of all or a substantial portion of our sales to any large-volume customer could have a material adverse effect on our financial condition and results of operations by reducing cash flows and our ability to spread costs over a larger revenue base".

Customer concentration offers suppliers benefits as well as constraints, arising from the power major customers hold over the supplier (Matsumura & Schloetzer, 2018; Wali Ullah et al., 2024). High customer concentration can adversely affect a supplier's profitability if major customers use their bargaining power to negotiate and enforce favourable contract terms at the supplier firm's expense (Dhaliwal et al., 2016; Gosman & Kohlbeck, 2009; Patatoukas, 2012). Conversely, supply chain collaborators can gain from each other by collaborating and coordinating their operations more efficiently, to ensure mutual benefits (Patatoukas, 2012; Schloetzer, 2012).

Based on the opportunities and risks associated with a supplier firm's major customers, we anticipate that a concentrated customer base will influence 10-K reports for several reasons. First, suppliers with high customer concentration experience heightened cash flow risks because losing a major customer can lead to a substantial decline in cash flow (Dhaliwal et al., 2016; Huang et al., 2016). These suppliers often invest significantly in relationship-specific assets that hold little value outside the current customer-supplier arrangement, resulting in the potential for considerable losses if they lose that major customer (Campello & Gao, 2017; Dong et al., 2021). Additionally, major customers leverage their bargaining power to secure preferential terms that can diminish profitability while increasing cash flow volatility for suppliers (Ghosh et al., 2006; Kouvelis, Wu, & Xiao, 2019; Piercy, 2009). Thus, to satisfy and maintain business relationships with major customers, supplier firms may engage in practices such as hoarding bad news (Chen et al., 2024), managing earnings upward (Raman & Shahrur, 2008) and concealing negative information about litigation loss contingencies (Cen et al., 2017). Managers seeking to conceal value-decreasing information and poor performance (Davis & Tama-Sweet, 2012; Jung et al., 2018), may lead to less readable 10-K reports (Hasan, 2020; F. Li, 2008).

Furthermore, suppliers with a concentrated customer base are likely to obfuscate their disclosures to balance power dynamics, manage risk and maintain price elasticity. As customers gain significant bargaining power over such suppliers, managers of these suppliers may become more motivated to obfuscate their financial reports to preserve or strengthen their position (Cheng et al., 2021; Leung et al., 2015). Doing so would likely give these suppliers control over the information flow during negotiations with major customers (Allender et al., 2021; Williamson, 1985). As major customers exert pressure on dependent suppliers for better terms or lower prices (Gosman & Kohlbeck, 2009), information obfuscation allows the supplier to hide cost information and so create information asymmetry (Ellison & Wolitzky, 2012; Grennan & Swanson, 2020).

Finally, linguistic complexity in corporate disclosures has two latent components: obfuscation and information content (Bushee et al., 2018). Consistent with this idea, the obfuscation motive to create information asymmetry in the customer-supplier relationship may lead suppliers to produce poor quality financial reports. However, even if we argue that a dependent supplier benefits from its concentrated customer base and does not intentionally create an asymmetric information environment, more information content of linguistic complexity resulting from overly technical disclosures (also referred to as information overload) could also produce less readable statements (Campbell et al., 2014; Lee, 2012). Moreover, if the supplier is facing financial difficulties because of its concentrated customer base, such firms may face pressure from their stakeholders to explain their predicament with more information (Bloomfield, 2008), which can increase the linguistic complexity and produce poor disclosure readability. Higher information content in disclosures and earnings calls may provide positive signals of future earnings growth, particularly for firms with greater earnings uncertainty, even if the disclosures are more complex (Aghamolla & Smith, 2023; Bushee et al., 2018; Bushee & Huang, 2024; Guo et al., 2021; Skinner, 2024). From this aspect, the added disclosure complexity has been found to reduce the readability of 10-K reports (Miller, 2010; You & Zhang, 2009). Based on the above arguments, we hypothesize that.

H1. Suppliers with high customer concentration produce less readable 10-K reports.

On the other hand, major customer firms that wield greater bargaining power over their dependent suppliers may incentivize closer monitoring of those suppliers. When a supplier relies heavily on its major customers, it must adhere to the customer demands and cultivate a closer relationship. To ensure compliance with these demands, customers are likely to monitor the activities of such suppliers and require them to provide clear disclosures (Anderson & Jap, 2005; Johnsen & Lacoste, 2016). This customer influence can also prompt suppliers to adopt more conservative reporting practices and enhance their audit quality (Hui et al., 2012; Krishnan et al., 2019). According to the certification hypothesis, major customers function as monitoring and certifying agents, thereby reducing information asymmetry for suppliers with high customer concentration, which can lead to lower cost of debt (Cai & Zhu, 2020; Cen et al., 2016). To maintain these important customer relationships, suppliers are likely to increase communication frequency and lower information costs (Cannon & Homburg, 2001). As major customers represent a significant portion of a supplier's revenue, monitoring efforts may intensify, resulting in improved corporate governance practices in supplier firms that help minimize opportunities for financial information distortion and reduce information asymmetry (Cai & Zhu, 2020; Cen et al., 2016; Do et al., 2023; Kelly & Gosman, 2000). Greater monitoring is also likely to force managers to move away from obfuscating disclosures (Chakrabarty et al., 2018). Studies show that greater monitoring through internal and institutional stakeholders is likely to decrease information asymmetry and improve the readability of disclosures (Choi & Chung, 2023; Ertugrul et al., 2017). Therefore, we hypothesize that the monitoring role played by major customers for suppliers with high customer concentration can improve the readability of their 10-K reports.

H2. Suppliers with high customer concentration produce more readable 10-K reports.

3. Data and methodology

3.1. Sample selection

We obtained accounting data from Compustat North America, customer-supplier data from the Compustat customer segment files

and the readability measures (the Bog Index and the 10-K document file size) from publicly available sources. Appendix 1 details how the study's variables were constructed. Our final sample includes 107,277 firm-year observations from 9554 non-financial firms based in the United States from 1994 to 2020. We initiated our sample in 1994, because 10-K documents became widely accessible in the EDGAR database around then. However, not all firms have reported major customers in the Compustat customer segment files because of requirements such as "SFAS No. 14 (before 1997) and SFAS No. 131 (after 1997)", which mandate that suppliers disclose the presence of and sales to any major customers contributing more than 10 % of their revenue, regardless of the number of segments they operate. Consequently, if we focus only on those firms that have identifiable major customers, our sample size is reduced to 20,004 firm-year observations, consisting of 3121 supplier firms.

3.2. Dependent variable: measures of annual report readability

Based on previous studies on annual report readability measures, we use the BOG Index and the Net File Size of 10-K documents for individual firms as the study's primary dependent variables (Bonsall IV et al., 2017; Hasan, 2020). The BOG Index assesses the quality of plain English writing as identified by linguistic experts and the SEC Plain English Handbook. It is calculated by summing the Sentence Bog (which highlights readability issues linked to sentence length) and the Word Bog (which quantifies plain English style problems and word difficulty, multiplied by 250 and divided by the total number of words), then subtracting the third component, Pep (which identifies writing features that enhance text understandability for readers). Higher BOG Index values indicate that a 10-K report is less readable.

Additionally, previous studies have used measures based on writing characteristics, such as document length (i.e., the total word count in written reports), to assess the readability of 10-K filings. The file size of 10-K documents has been quantified as the number of megabytes in the entire filing as recorded in the EDGAR filing system. We take the natural logarithm of the file size (File Size) as an alternative readability measure. Although this approach has been critiqued for being "noisy" in capturing aspects beyond simple quantity-based readability, it has been extensively used (Bonsall IV et al., 2017; Hasan, 2020; Loughran & McDonald, 2014; You & Zhang, 2009) and adds robustness to our findings using the BOG Index.

Although the Fog index has been frequently used in previous studies (De Franco et al., 2015; Lawrence, 2013; F. Li, 2008; Miller, 2010), it is often criticized for being inadequately specified for assessing the readability of financial documents. The Fog index consists of two components: average words per sentence and the proportion of complex words. Business texts typically feature a high frequency of complex words (defined as those with more than two syllables), which generally reduces readability. For instance, common terms found in 10-K reports, such as "corporation," "agreement," and "telecommunications," are classified as complex words, yet they are easily understood by major corporate customers, analysts, and other stakeholders. Furthermore, the Fog index does not effectively capture the various dimension attributes emphasized by the SEC's plain English rule. As a result, the complex words component of the Fog Index introduces measurement error with minimal predictive power regarding accounting and finance issues, such as post-filing stock return volatility and analyst forecast dispersion (Bonsall IV et al., 2017; Loughran & McDonald, 2014; Rjiba et al., 2021). Therefore, we opted not to use the Fog index as our primary measure of 10-K readability. However, we consider it an alternative proxy to enhance the robustness of our findings.

3.3. Independent variables: measures of customer concentration

Following Dhaliwal et al. (2016), we use three proxies to measure a supplier's degree of customer concentration. First, we identify a supplier firm's major customers using Compustat's segment customer files. Since 1976, the Financial Accounting Standards Board (FASB) has required, under SFAS 14, that suppliers disclose external clients that contribute 10 % or more of their revenue. In 1997, FAS 131 replaced SFAS 14 and, under SEC Regulation S-K Item 101, public firms are still mandated to identify these significant customers. Although suppliers are required to disclose customers generating at least 10 % of revenue, many also voluntarily list customers that contribute less than this threshold. We exclude these voluntarily disclosed customers from our calculations of customer concentration because their inclusion could lead to potential conflicts of interest. For our initial measure of customer concentration, we create a dummy variable (Major Customer) that equals one if a supplier identifies at least one corporate customer that accounts for 10 %, or more, of its annual revenue and zero otherwise (Dhaliwal et al., 2016; Huang et al., 2016). For our secondary measure, we use the Herfindahl-Hirschman Index (Major Customer HHI) to quantify major customer concentration (Dhaliwal et al., 2016; Patatoukas, 2012). This index accounts for both the number of major customers reported by the supplier and their relative importance to the supplier's overall annual sales. Consequently, we calculate the customer concentration for supplier i in year t across its J major clients as:

$$Major\ Customer\ HHI_{it} = \sum_{j=1}^{J} \left(\frac{Sales_{ijt}}{Sales_{it}} \right)^2 \tag{1}$$

where: $Sales_{it}$ denotes a supplier's total sales in year t and $Sales_{ijt}$ denotes a supplier's sales to a large client j in year t. A value between 0 and 1 represents the Major Customer HHI, with higher values indicating a more concentrated customer base. This metric equals zero for suppliers who do not disclose sales to a major customer and one for firms whose entire annual revenue depends on a single major customer.

Our third measure is Major Customer Sales, calculated as the portion of a supplier's annual total sales that comes from all customers who contribute at least 10 % of that supplier's annual revenue (Banerjee et al., 2008; Dhaliwal et al., 2016).

Table 1
Summary statistics.

Variable	Obs	Mean	SD	Min	25th Percentile	Median	75th Percentile	Max
Bog Index	107,277	82.90	8.48	70.00	77.00	83.00	88.00	97.00
Ln (File Size)	79,899	12.68	0.57	10.18	12.33	12.72	13.08	13.61
Major Customer Sales	107,277	0.04	0.13	0.00	0.00	0.00	0.00	0.78
Major Customer HHI	107,277	0.02	0.08	0.00	0.00	0.00	0.00	0.62
ROA	107,277	0.02	0.38	−1.36	−0.01	0.07	0.12	43.88
Leverage	107,277	0.26	0.26	0.00	0.04	0.18	0.40	0.91
Book to Market Ratio	107,277	0.53	0.56	−0.58	0.19	0.46	0.82	1.84
Size	107,277	5.80	2.66	0.00	3.99	6.04	7.60	15.2
Intangible Assets	107,277	0.09	0.15	0.00	0.00	0.02	0.11	1.00
Asset Tangibility	107,277	0.18	0.22	0.00	0.02	0.09	0.29	1.00
Cash Flow to Assets	107,277	0.16	0.22	0.00	0.02	0.08	0.21	1.00
Geographic Segments	107,277	3.87	5.05	1.00	1.00	2.00	4.00	152.00
Business Segments	107,277	3.90	6.01	1.00	1.00	1.00	3.00	122.00
Firm Age	107,277	21.01	30.59	1.00	4.00	9.00	23.00	37.00
Free Cash Flow	107,277	−0.07	0.29	−14.60	−0.08	0.01	0.05	6.06

Note: This table presents the summary statistics results. Obs = Number of observations, SD= Standard deviation, Min = Minimum and Max = maximum.

3.4. Research model

We estimate the following regression model to examine the impact of customer concentration on the readability of annual 10-K reports:

$$Readability_{i,t} = \beta_0 + \beta_1 Customer\ Concentration_{i,t} + Controls_{i,t} + YEAR\ FE + INDUSTRY\ FE + \varepsilon_{i,t}$$

where: the BOG and File Size serve as proxies for the dependent variable, the 10-K report's readability. Our primary variable of interest is Customer Concentration, measured by three concentration proxies. Following previous studies, we include control variables that are associated with the readability of 10-K reports and customer concentration: ROA (the profitability of the firm), Intangible Assets, Size (the natural logarithm of the book value of firm assets), Leverage, Book to Market Value Ratio, Asset Tangibility, Cash Flow to Assets, Firm Age (the natural logarithm of the age of the firm's operations), Free Cash Flow and the number of Geographic and Business Segments of the firm. Details on how these controls are measured are provided in Appendix Table 1. Additional governance related variables (board size, board gender diversity and board independence) are included as controls, with the results reported in Appendix Table 3. We also considered mean analyst following as a control variable, but its inclusion shrinks our sample size significantly.[5]

3.5. Summary statistics

Table 1 presents the summary statistics of all variables. They show that the BOG index has a mean of 82.90 with a standard deviation of 8.48, indicating that, on average, our sample firms have poor readability 10-K reports, which aligns with previous studies (Bonsall & Miller, 2017; Hasan, 2020). The descriptive statistics of the three customer concentration measures indicate substantial variation across the dataset. Among the control variables, most exhibit high variations in mean and standard deviation, which is expected because our sample comprises 9554 firms from various industrial sectors. Overall, our summary statistics are comparable to prior studies (Dhaliwal et al., 2016; Hasan, 2020).

4. Empirical findings

4.1. Baseline regression results

We begin our analysis by testing the hypothesized relationship between 10-K report readability and customer concentration. Table 2 presents the results from our baseline regression models for all suppliers, i.e., those with and without major customers. We conducted OLS regression analyses incorporating year and industry-fixed effects. Models 1–3 examine the relationship between the three measures of customer concentration and the BOG Index, and models 4–6 focus on file size. The results from models 1–6 reveal a positive, statistically significant coefficient for all customer concentration proxies in relation to 10-K report readability. Given that a higher BOG Index and file size indicate lower readability, the findings suggest that suppliers with a concentrated customer base produce 10-K reports that are more challenging to read, thus supporting our proposed negative relationship between customer concentration and the readability of 10-K reports. For the control variables, the coefficients for firm size, ROA, intangible assets, cash flow to assets, geographic and business segments, and firm age are consistent across all models. In contrast, the coefficients for the book-to-market ratio, leverage, and asset tangibility show variation, but they are insignificant in these instances.

[5] Results are unreported but available upon request.

Table 2
Baseline regression results – observations for all suppliers.

Dependent Variable	Bog Index			Ln (File Size)		
	(1)	(2)	(3)	(4)	(5)	(6)
Major Customer Sales	1.9562***			0.1285***		
	(0.000)			(0.000)		
Major Customer HHI		2.7164***			0.1794***	
		(0.000)			(0.000)	
Major Customer			0.8428***			0.0580***
			(0.000)			(0.000)
Leverage	−0.0005*	−0.0005*	−0.0005*	0.0003	0.0003	0.0003
	(0.071)	(0.071)	(0.071)	(0.277)	(0.280)	(0.272)
Book to Market Ratio	−0.0004	−0.0006	−0.0004	0.0003	0.0003	0.0003
	(0.127)	(0.125)	(0.138)	(0.876)	(0.874)	(0.879)
Size	1.0780***	1.0798***	1.0750***	0.1444***	0.1445***	0.1442***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ROA	−2.9588***	−2.9428***	−2.9607***	−0.2191***	−0.2181***	−0.2192***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Intangible Assets	3.7971***	3.8147***	3.7543***	0.2791***	0.2804***	0.2757***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Asset Tangibility	−0.6170***	−0.6028***	−0.6362***	0.0111	0.0118	0.0095
	(0.000)	(0.000)	(0.000)	(0.249)	(0.217)	(0.318)
Cash Flow to Assets	5.5562***	5.5708***	5.5672***	0.1613***	0.1621***	0.1621***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Geographic Segments	0.0586***	0.0590***	0.0569***	0.0013***	0.0013***	0.0012**
	(0.000)	(0.000)	(0.000)	(0.009)	(0.008)	(0.017)
Business Segments	−0.0565***	−0.0560***	−0.0573***	0.0013***	0.0014***	0.0013***
	(0.000)	(0.000)	(0.000)	(0.002)	(0.001)	(0.002)
Ln (Firm Age)	−0.5439***	−0.5449***	−0.5467***	−0.0466***	−0.0466***	−0.0470***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Free Cash Flow	−0.0042	−0.0032	−0.0105	−0.1753***	−0.1750***	−0.1760***
	(0.974)	(0.980)	(0.935)	(0.000)	(0.000)	(0.000)
Constant	77.6487***	77.6658***	77.6368***	11.8608***	11.8624***	11.8592***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Observations	107,277	107,277	107,277	79,899	79,899	79,899
Adjusted R-squared	0.4111	0.4108	0.4114	0.4470	0.4467	0.4474
Mean VIF	2.24	2.23	2.24	2.16	2.17	2.15
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES

Note: The baseline regression results for major customer concentration and financial statement readability are presented in this table. In models 1–3, the dependent variable is the *Bog Index*. In models 4–6 *File Size* is the dependent variable. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

Our analysis reported in Table 2 considers suppliers with and without major customers. To ensure that our identified relationship is robust, we re-estimate our baseline models for a sample comprising only suppliers with identified major customers. Table 3 reports those results. As in Table 2, customer concentration proxies have positive coefficients for the 10-K report readability proxies. These results are economically significant. For example, a 1 % increase in the Major Customer Sales proxy leads to a 0.62 increase in the BOG Index and a 4.01 % increase in the file size of the supplier's 10-K reports, holding all other factors constant. Considering the mean file size of our sample is 0.37 megabytes, a 1 % increase in the Major Customer Sales proxy leads to a 10.83 % (=0.0401/0.37) increase in the file size from the mean, indicating its economic significance. Similarly, for the BOG Index, such an increase in Major Customer Sales is equivalent to a 0.74 % (=0.6165/82.90) increase from the mean. For the same increase in the Major Customer HHI proxy, the BOG Index and file size of our sample increased by 1.11 % (=0.9217/82.86) and 19.19 % (=0.0710/0.37), respectively. We also use firm fixed effects to show that this relationship is not driven by omitted time-invariant factors. The results using firm fixed effects in Table 4 show quantitatively similar results, confirming that our findings are not driven by firm-level unobserved heterogeneity. Further, the mean variance inflation factor (VIF) for the baseline regressions reported in Table 2 also shows a lack of substantial multicollinearity adversely affecting our findings. These results corroborate our baseline findings, demonstrating that even for suppliers with identified major customers, a concentrated customer base is associated with declining readability of 10-K reports.

We recognize that the economic impact of customer concentration proxies on the BOG Index appears to be modest. However, on closer inspection, we note that the BOG Index itself exhibits limited variation, with an interquartile range of merely 11. According to the readability classifications based on the BOG Index, firms at both the 25th and 75th percentiles fall into the "Poor" readability category. Consequently, although the coefficients are statistically significant, the minimal interquartile dispersion in this measure

Table 3
Baseline regression results – observations only for suppliers with major customers.

Dependent Variable	Bog Index		Ln (File Size)	
	(1)	(2)	(3)	(4)
Major Customer Sales	0.6165***		0.0401***	
	(0.001)		(0.007)	
Major Customer HHI		0.9217***		0.0710***
		(0.000)		(0.000)
Leverage	0.2806***	0.2771***	0.0399***	0.0396***
	(0.000)	(0.000)	(0.000)	(0.000)
Book to Market Ratio	−0.0001*	−0.0001*	−0.0001	−0.0001
	(0.060)	(0.060)	(0.182)	(0.181)
Size	1.0567***	1.0573***	0.1449***	0.1449***
	(0.000)	(0.000)	(0.000)	(0.000)
ROA	−5.1271***	−5.1309***	−0.1780***	−0.1791***
	(0.000)	(0.000)	(0.000)	(0.000)
Intangible Assets	4.7251***	4.7291***	0.2671***	0.2674***
	(0.000)	(0.000)	(0.000)	(0.000)
Asset Tangibility	0.3945	0.3853	−0.0067	−0.0075
	(0.153)	(0.162)	(0.773)	(0.747)
Cash Flow to Assets	6.8756***	6.8443***	0.1564***	0.1528***
	(0.000)	(0.000)	(0.000)	(0.000)
Geographic Segments	0.0641***	0.0645***	0.0015	0.0015
	(0.000)	(0.000)	(0.171)	(0.157)
Business Segments	−0.0443***	−0.0441***	0.0007	0.0007
	(0.000)	(0.000)	(0.445)	(0.438)
Ln (Firm Age)	−0.6181***	−0.6165***	−0.0491***	−0.0489***
	(0.000)	(0.000)	(0.000)	(0.000)
Free Cash Flow	0.6931*	0.6934*	−0.2191***	−0.2186***
	(0.093)	(0.092)	(0.000)	(0.000)
Constant	77.5990***	77.6509***	11.8882***	11.8899***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	20,044	20,044	17,005	17,005
Adjusted R-squared	0.4122	0.4123	0.4275	0.4277
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: This table presents the results of the baseline regression for major customer concentration and financial statement readability with observations only for suppliers with major customers. In models 1 and 2, the dependent variable is the *Bog Index*. Models 3 and 4 have *File Size* as the dependent variable. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parentheses. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

results in a relatively lower economic significance of customer concentration proxies in diminishing the BOG Index. These observations align with previous research using the same readability proxy (Habib & Hasan, 2020; Hasan, 2020).

Prior studies highlight that various board governance mechanisms, such as board gender diversity, board independence, and board size, influence the readability of financial reports (Nadeem, 2022; Rahman & Kabir, 2023). To address potential concerns regarding the exclusion of these board-related factors from our analysis, we incorporate them as additional control variables. Our results remain stable after accounting for these board governance-related controls (board independence, board gender diversity, and board size), with the results detailed in Appendix Table 3. Though the Fog Index is frequently used as an alternative measure of 10-K report readability, we elaborated in Section 3.2 on its inadequacies in our research context. Nevertheless, we re-estimated our baseline regressions using the Fog Index as the readability measure and obtained comparable results, which are presented in Appendix Table 4.

4.2. Mitigating endogeneity concerns

Our preliminary findings from the baseline and robust subsample tests support the idea that customer concentration decreases 10-K report readability. These findings, however, may not reflect a causal association between major customer concentration and financial statement readability, since they could be explained by omitted firm characteristics or variables. To address this potential endogeneity issue, we conduct a two-stage least squares (2SLS) regression analysis with instrumental variables, Propensity Score Matching (PSM), Entropy Balancing, and the Oster (2019) test.

Table 4
Baseline regression results – inclusion of firm fixed effects.

Dependent Variable	Bog Index			Ln (File Size)		
	(1)	(2)	(3)	(4)	(5)	(6)
Major Customer Sales	0.0829**			0.0603***		
	(0.049)			(0.000)		
Major Customer HHI		0.0493*			0.0818***	
		(0.078)			(0.000)	
Major Customer			0.0628*			0.0241***
			(0.076)			(0.000)
Leverage	−0.0006***	−0.0005***	−0.0006***	−0.0001	−0.0001	−0.0001
	(0.005)	(0.005)	(0.000)	(0.692)	(0.694)	(0.692)
Book to Market Ratio	−0.0001	−0.0001	−0.0001	−0.0004	−0.0004	−0.0004
	(0.675)	(0.875)	(0.748)	(0.145)	(0.135)	(0.148)
Size	0.7188***	0.7190***	0.7183***	0.0932***	0.0933***	0.0932***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ROA	−2.1620***	−2.1613***	−2.1623***	−0.1190***	−0.1192***	−0.1187***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Intangible Assets	0.9743***	0.9734***	0.9742***	0.0809***	0.0810***	0.0802***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Asset Tangibility	−0.4320***	−0.4314***	−0.4327***	0.0008	0.0006	0.0011
	(0.003)	(0.000)	(0.003)	(0.964)	(0.961)	(0.949)
Cash Flow to Assets	−0.1822	−0.1816	−0.1813	−0.0874***	−0.0875***	−0.0864***
	(0.107)	(0.109)	(0.121)	(0.000)	(0.000)	(0.000)
Geographic Segments	0.0044	0.0044	0.0042	0.0020***	0.0021***	0.0020***
	(0.601)	(0.620)	(0.603)	(0.026)	(0.028)	(0.027)
Business Segments	−0.0322***	−0.0321***	−0.0325***	0.0027***	0.0028***	0.0027***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Ln (Firm Age)	−0.6664***	−0.6667***	−0.6663***	−0.0004	−0.0005	−0.0004
	(0.000)	(0.000)	(0.000)	(0.913)	(0.897)	(0.903)
Free Cash Flow	0.3246***	0.3244***	0.3248***	0.0558***	0.0557***	0.0558***
	(0.002)	(0.000)	(0.002)	(0.000)	(0.000)	(0.000)
Constant	81.3784***	81.3807***	81.3747***	12.1049***	12.1059***	12.1040***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Observations	107,277	107,277	107,277	79,899	79,899	79,899
Adjusted R-squared	0.7655	0.7927	0.7627	0.6661	0.6660	0.6664
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES

Note: The baseline regression results with the inclusion of firm fixed effects are presented in this table. In models 1 to 3, the dependent variable is the *Bog Index*. Models 4 to 6 have *File Size* as the dependent variable. Year and firm fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

4.2.1. 2SLS regression using instrumental variables

In this section, we use 2SLS regression analysis to alleviate endogeneity concerns that could result from omitted variables. We consider an adverse customer industry shock as our instrumental variable.[6] We posit this to be a credible instrument because industry shocks to customers are potentially exogenous to supplier firms and less likely to affect them through channels apart from the supply chain relationships, satisfying the exclusion criterion for a valid instrument. Such negative industry shocks can reduce customer demand and create potential instability through customer exits or mergers and acquisitions, satisfying the relevance criterion. Consequently, suppliers with a concentrated customer base are expected to experience significant negative impacts from adverse shocks in the industries of their customers (Albuquerque et al., 2014; Bhattacharyya & Nain, 2011; Ma et al., 2020). Consistent with previous research, we define an adverse industry shock as a decline of 10 % or more in a customer's annual stock return. We create an indicator variable that is set to one for all major customers if their industry experiences an annual stock return drop of −10 % or lower, and zero otherwise. We then compute the variable BadCustIndShock as the weighted average of these indicator variables, considering the percentage of sales attributed to each corresponding major customer (Albuquerque et al., 2014; Ma et al., 2020).

The results of the 2SLS regressions are detailed in Table 5. Panel A and B presents the results for the first and second stage, respectively. The first-stage results reveal that the BadCustIndShock instrument is statistically significant for customer concentration proxies (Major Customer Sales and Major Customer HHI). This finding indicates that negative shocks to customer industries lead to increased corporate customer concentration for upstream suppliers, as anticipated. The robustness of this instrument is confirmed through a series of diagnostic tests, including the Anderson under identification test and the Cragg-Donald weak instrument test. The

[6] We thank the anonymous reviewer for this suggestion.

Table 5
2SLS (IV – Adverse shock to customer industries).

Panel-A: First Stage Regressions

Dependent Variable	(1)	(2)
	Industry Major Customer Sales	Industry Major Customer HHI
BadCustIndShock	0.0611***	0.0287***
	(0.000)	(0.000)
Leverage	0.0047	0.0073**
	(0.201)	(0.017)
Book to Market Ratio	0.0002**	0.0002***
	(0.023)	(0.006)
Size	−0.0061***	−0.0040***
	(0.000)	(0.000)
ROA	0.0618***	0.0386**
	(0.003)	(0.019)
Intangible Assets	−0.0287**	−0.0268***
	(0.015)	(0.002)
Asset Tangibility	0.0234**	0.0241***
	(0.047)	(0.008)
Cash Flow to Assets	0.1172***	0.1116***
	(0.000)	(0.000)
Geographic Segments	−0.0053***	−0.0039***
	(0.000)	(0.000)
Business Segments	0.0002	−0.0007
	(0.824)	(0.161)
Ln (Firm Age)	−0.0119***	−0.0095***
	(0.000)	(0.000)
Free Cash Flow	−0.0197	−0.0130
	(0.281)	(0.375)
Constant	0.1854***	0.0522**
	(0.000)	(0.037)
Observations	107,277	107,277
Adjusted R-squared	0.0969	0.1108
Year FE	YES	YES
Industry FE	YES	YES
Underidentification Test (p-value)	1294.1512 (0.000)	1524.3012 (0.000)
Cragg-Donald Wald F Statistic	1296.3102	1521.3712

Panel-B: Second Stage Regressions – Dependent Variables: ***Bog Index*** and ***File Size***

Dependent Variable	(1)	(2)	(3)	(4)
	Bog Index		Ln (File Size)	
Predicted Major Customer Sales	9.4697***		0.5781***	
	(0.000)		(0.000)	
Predicted Major Customer HHI		23.3003***		1.3270***
		(0.000)		(0.000)
Leverage	0.5800***	0.4123***	0.0630***	0.0530***
	(0.000)	(0.000)	(0.000)	(0.000)
Book to Market Ratio	0.0006***	0.0001	0.0002***	0.0001***
	(0.006)	(0.616)	(0.000)	(0.000)
Size	1.1947***	1.2108***	0.1565***	0.1559***
	(0.000)	(0.000)	(0.000)	(0.000)
ROA	−9.2366***	−9.3613***	−0.4708***	−0.4802***
	(0.000)	(0.000)	(0.000)	(0.000)
Intangible Assets	7.3689***	7.6060***	0.4944***	0.5007***
	(0.000)	(0.000)	(0.000)	(0.000)
Asset Tangibility	−0.5506*	−1.0647***	0.1250***	0.0972***
	(0.054)	(0.001)	(0.000)	(0.000)
Cash Flow to Assets	10.3168***	8.3432***	0.3293***	0.2164***
	(0.000)	(0.000)	(0.000)	(0.000)
Geographic Segments	0.0403*	0.0951***	−0.0117***	−0.0089***
	(0.097)	(0.000)	(0.000)	(0.000)
Business Segments	0.0850***	0.0832***	0.0187***	0.0190***
	(0.000)	(0.000)	(0.000)	(0.000)
Ln (Firm Age)	−0.1969***	−0.1408***	−0.0173***	−0.0137***
	(0.000)	(0.002)	(0.000)	(0.000)
Free Cash Flow	2.9977***	3.0485***	−0.0846	−0.0767

(*continued on next page*)

Table 5 (*continued*)

Panel-B: Second Stage Regressions – Dependent Variables: ***Bog Index*** and ***File Size***				
Dependent Variable	(1)	(2)	(3)	(4)
	Bog Index		Ln (File Size)	
	(0.000)	(0.000)	(0.158)	(0.201)
Constant	72.2758***	71.9326***	11.4780***	11.4773***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	107,277	107,277	93,097	93,097
R-squared	0.1763	0.1842	0.2987	0.2068
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: The findings of a two-stage lease-squares regression analysis using the instrumental variable of adverse shock to customer industries (Bad industry shock) are shown in this table. The findings of the first stage OLS regression analysis are presented in Panel A and second stage models are presented in Panel B. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

test statistics demonstrate that the instrument is properly identified, rejecting the null hypotheses of being irrelevant or weak, and confirming that it is exogenous to the readability of a supplier's 10-K report. The estimates from the second-stage regression reveal positive, statistically significant coefficients between the instrumented major customer, major customer sales and the HHI values in relation to the BOG Index and file size, reinforcing the validity of these instruments that align with our baseline findings.[7]

4.2.2. Propensity score matched (PSM) and entropy balanced sample

Our linear model regressing customer concentration against 10-K report readability may capture some nonlinear effects that may not be fully accounted for between suppliers with concentrated clients and those with diversified clients. Apart from this potential model misspecification issue, there may be some sample selection bias arising from significant differences between suppliers with and without major customers. As in prior literature (Dhaliwal et al., 2016; Fang et al., 2014; Lee et al., 2020; Wen, Ke, & Liu, 2021), we use a propensity score matching (PSM) approach to mitigate sample selection bias that may arise from the potential endogenous selection of observed variables and to address any concerns regarding model misspecification. To establish the PSM sample, we first estimate a logit model using the complete dataset. This allows us to categorize each supplier with at least one identified major customer (the treatment group) and match them with a control firm that has no major customers in the same year (the control group). As a result, we create a sample consisting of 20,026 observations for the treatment group and 20,026 for the control group.

We generate a propensity score for each firm based on all the controls included in our baseline regression model. Subsequently, we select the control firm with the closest propensity score to each treatment firm. Table 6, Panel A, presents the mean differences in observable characteristics, revealing that there are no significant differences for most control variables. We then estimate our baseline model using the PSM-matched sample. The findings in Table 6, Panel B, indicate positive, consistently statistically significant coefficients for the customer concentration measures across the various proxies for 10-K report readability. These results from the PSM subsample further enrich and validate our conclusion that greater customer concentration among suppliers results in decreased readability of their 10-K reports.

Though the PSM approach is widely used in finance and accounting research, it has been criticized for the substantial loss of observations. The entropy balancing approach has recently gained popularity as an innovative method to address substantial differences in observable covariates between the treatment and control groups without significantly reducing the sample size (Al-Hadi et al., 2022; Amiram et al., 2017; Chapman & Green, 2018; Zaman, 2023). This method achieves a much better covariate balance than PSM by considering mean, variance, and skewness, all while preserving crucial information in the data. Inspired by earlier research (Bose et al., 2021; Hainmueller, 2012; Hossain et al., 2022), we apply the entropy balancing method to create our samples and present the multivariate regression findings in Table 7.

We initiate the entropy balancing process by ensuring that the variables are balanced across three dimensions: mean, skewness and variance. Firm-year observations for companies with major customers are categorized into the treatment group and those without major customers are placed in the control group. The results demonstrating the proof of convergence for entropy balancing are presented in Table 7, Panel A, and the regression analysis results using the entropy-balanced matched samples are shown in Table 7, Panel B. Our findings indicate that the positive, statistically significant relationship between customer concentration measures and 10-K report readability remains intact. These findings further validate our primary results.

[7] Following past studies, we also conducted 2SLS analysis using two- and three-year lagged average industry customer concentration measurements as instrumental variables (Dhaliwal et al., 2016; Lee et al., 2020). Results reported in Appendix Table 5 show that our findings remain consistent.

Table 6
Propensity score matching results (PSM) (matching based on suppliers with and without major customers).

Panel – A (Differences in observables)			
	Group (Major Customer = 1) Treated	Control Group (Major Customer = 0) Control	
	Mean	Mean	t-statistic
Dependent Variable			
Bog Index	84.05	82.91	15.81
Ln (File Size)	12.70	12.65	7.69
Control Variable			
Leverage	0.29	0.30	−1.83
Book to Market Ratio	15.21	10.45	−1.02
Size	5.54	5.52	1.06
ROA	0.06	0.07	−6.43
Intangible Assets	0.12	0.13	−2.76*
Asset Tangibility	0.25	0.26	−4.66
Cash Flow to Assets	0.03	0.04	−3.12*
Geographic Segments	6.39	6.06	4.81
Business Segments	6.70	6.29	5.17
Ln (Firm Age)	2.38	2.33	3.39
Free Cash Flow	−0.04	−0.03	−3.27

Panel – B (Multivariate results using matched samples)				
Dependent Variable	Bog Index		File Size	
	(1)	(2)	(3)	(4)
Major Customer Sales	2.4267***		0.1192***	
	(0.000)		(0.000)	
Major Customer HHI		3.2698***		0.1576***
		(0.000)		(0.000)
Constant	79.2517***	79.3663***	11.9004***	11.9074***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	40,052	40,052	32,328	32,328
Control Variables	YES	YES	YES	YES
Adjusted R-squared	0.3933	0.3925	0.4359	0.4355
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: Panel A displays the results of the differences in observables for PSM. Panel B displays the results of the findings for the PSM matched sample. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

4.2.3. The Oster (2019) test for omitted variable bias

Though our baseline regression findings are robust to the inclusion of additional governance-related control variables, there is a possibility that our model may not encompass all relevant controls. That could lead to omitted variable bias, providing a biased estimate of the coefficients. We use the bound estimates technique to address such concerns (Oster, 2019). The Oster (2019) test evaluates the stability of regression coefficients by analysing the effects of adding and removing controls, alongside the corresponding R-squared values, to generate a new identifiable parameter. If the identifiable set produced does not include a 0 value, it suggests that the estimate is unlikely to be significantly impacted by omitted variable bias. The method has been used in several recent studies in finance and accounting to address endogeneity concerns related to omitted variables (Hasan et al., 2024; Hasan & Uddin, 2022; Jha et al., 2024).

Based on prior literature (Hasan et al., 2024; Oster, 2019), we set $\delta = 1$ and $R_{max} = \min(1.3\,\widetilde{R,1})$ and provide the results in Table 8 which demonstrate that the estimated identified set for the Bog index does not contain a 0, suggesting a lack of omitted variable bias disrupting our baseline results. We find consistent outcomes for the alternative readability proxy, Ln (File Size). The results show our inference remains consistent if we use $R_{max} = \min(1.5\,\widetilde{R,1})$ or $R_{max} = \min(2.2\,\widetilde{R,1})$.

5. Additional analyses

5.1. Managerial linguistic complexity: obfuscation and information content

Our results so far suggest that suppliers with concentrated customers have reduced readability of 10-k reports. In the next analysis

Table 7
Entropy balance matched results (matching based on suppliers with and without major customers).

Panel - A (Proof of Convergence)						
Before Balancing						
Variable	Treatment (Major Customer = 1)			Control (Major Customer = 0)		
	Mean	Variance	Skewness	Mean	Variance	Skewness
Leverage	0.2464	0.0579	1.0714	0.2889	0.0772	0.9749
Book to Market Ratio	0.5275	0.2831	0.5345	0.4767	0.3412	0.4896
Size	5.5880	4.5162	−0.0145	5.1953	7.6690	0.0076
ROA	0.0654	0.0429	2.4172	0.0067	0.0667	−2.0842
Intangible Assets	0.1231	0.0247	1.6783	0.1148	0.0283	1.9664
Asset Tangibility	0.2570	0.0553	1.1682	0.2321	0.0557	1.1662
Cash Flow to Assets	0.1921	0.0486	1.4563	0.1636	0.0477	1.9742
Geographic Segments	5.5622	19.8002	0.7812	3.9884	14.7121	1.4712
Business Segments	5.7311	27.7602	0.8449	3.8312	20.2602	1.6781
Ln (Firm Age)	2.3333	1.8574	0.1315	2.2532	1.8753	0.2451
Free Cash Flow	−0.03157	0.0520	−0.7563	−0.0851	0.0969	−6.9743
After Balancing						
Variable	**Treatment** (Major Customer = 1)			**Control** (Major Customer = 0)		
	Mean	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**
Leverage	0.2464	0.0579	1.0714	0.2464	0.0579	1.0714
Book to Market Ratio	0.5275	0.2831	0.5345	0.5275	0.2831	0.5345
Size	5.5880	4.5162	−0.0145	5.5880	4.5162	−0.0145
ROA	0.0654	0.0429	2.4172	0.0654	0.0429	2.4172
Intangible Assets	0.1231	0.0247	1.6783	0.1231	0.0247	1.6783
Asset Tangibility	0.2570	0.0553	1.1682	0.2570	0.0553	1.1682
Cash Flow to Assets	0.1921	0.0486	1.4563	0.1921	0.0486	1.4563
Geographic Segments	5.5622	19.8002	0.7812	5.5622	19.8002	0.7812
Business Segments	5.7311	27.7602	0.8449	5.7311	27.7602	0.8449
Ln (Firm Age)	2.3333	1.8574	0.1315	2.3333	1.8574	0.1315
Free Cash Flow	−0.03157	0.0520	−0.7563	−0.03157	0.0520	−0.7563

Panel – B (Multivariate results using matched samples)				
Dependent Variable	Bog Index		File Size	
	(1)	(2)	(3)	(4)
Major Customer Sales	1.4565***		0.0834***	
	(0.000)		(0.000)	
Major Customer HHI		1.9293***		0.1126***
		(0.000)		(0.000)
Constant	77.5894***	77.6669***	11.8707***	11.8752***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	107,240	107,240	79,867	79,867
Control Variables	YES	YES	YES	YES
Adjusted R-squared	0.4004	0.4000	0.4228	0.4220
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: Panel A displays the findings for the proof of convergence before and after entropy balancing. Panel B displays the results of the findings for the entropy balance matched sample. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

we examine whether this is due to obfuscation or information content as discussed in section 2.3. Prior research has decomposed the linguistic complexity of 10-K reports into two latent components: the level of obfuscation and the information content, based on earnings conference call transcripts. The obfuscation component acts as a proxy for the use of unnecessarily complex words to reduce the understandability of disclosures. The information component refers to the amount of informative technical disclosure made by managers during conference calls (Bushee et al., 2018; Bushee & Huang, 2024; Gu et al., 2024; Hooghiemstra et al., 2017). Based on our literature review, we posit that managers of supplier firms with a concentrated customer base may be incentivized to conceal potential firm-related bad news to protect their interests. Managers motivated to hide bad news would likely do so by obfuscating disclosures and making them less readable. However, even if we argue that a dependent supplier benefits from its concentrated customer base and does not intentionally create an asymmetric information environment, higher information content of linguistic complexity resulting from overly technical disclosures (also referred to as information overload) could also produce less readable statements (Campbell et al., 2014; Lee, 2012). Using the following regression model, we identify whether the relationship between customer concentration and 10-K report readability is moderated and influenced by obfuscation or information content of managerial

Table 8
Oster (2019) test for omitted variable bias.

		(1)	(2)	(3)	(4)
Dependent variable	**Treatment Variable**	**Uncontrolled effect (P-value), [R^2]**	**Controlled effect (P-value), [R^2]**	**Identified set** $[\tilde{\beta}, \beta^*(min\{1.3\tilde{R}, 1\}, 1)]$	$\bar{\delta}$ **for** $\beta = 0$ **and** $R_{max} = 1$
Bog Index	Major Customer Sales	3.2850*** (0.0000) [0.0036]	1.9307*** (0.0000) [0.4128]	[0.4420, 1.2660]	2.3375
Bog Index	Major Customer HHI	5.0549*** (0.0000) [0.0033]	2.6989*** (0.0000) [0.4126]	[0.6254, 4.3000]	1.2685
File Size	Major Customer Sales	0.0625*** (0.0000) [0.0002]	0.1269*** (0.0000) [0.4126]	[0.13212, 10.8861]	12.6103
File Size	Major Customer HHI	0.0918*** (0.0000) [0.0002]	0.1780*** (0.0000) [0.4436]	[0.17643, 9.5506]	10.88614

Table 9
Moderation analysis – managerial linguistic complexity (obfuscation and information content) during conference calls.

Dependent Variable	(1)	(2)	(3)	(4)
	Bog Index		Ln (File Size)	
Major Customer HHI * Obfuscation (Both)	0.5697** (0.032)		0.0345** (0.044)	
Major Customer HHI * Information (Both)		0.2495 (0.308)		0.0078 (0.807)
Obfuscation (Both)	0.6399*** (0.000)		0.0286*** (0.000)	
Information (Both)		0.1377*** (0.000)		0.0052 (0.217)
Major Customer HHI	1.9157*** (0.000)	0.3802 (0.103)	0.0615*** (0.000)	0.0747*** (0.004)
Constant	82.5822*** (0.000)	83.2093*** (0.000)	12.1822*** (0.000)	12.1832*** (0.000)
Observations	24,406	65,677	24,161	51,430
Adjusted R-squared	0.3407	0.2912	0.3653	0.3467
Controls	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: The moderating effect of managerial linguistic complexity, in the form of the obfuscation and information content during conference calls, was examined and the results are presented in this table. In models 1 and 2, the dependent variable is the Bog Index. In models 3 and 4, File Size is the dependent variable. For brevity, only the Major Customer HHI is considered as the proxy for customer concentration. Our results remain similar when Major Customer Sales is considered as the customer concentration proxy and when we consider the two components of obfuscation and information content individually. Year and firm fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

linguistic complexity:

$$Readability_{i,t} = \beta_0 + \beta_1 Linguistic\ Complexity\ (Obfuscation\ or\ Information)_{i,t} + \beta_1 Customer\ Concentration_{i,t} + \beta_1 Customer\ Concentration_{i,t} \times Linguistic\ Complexity\ (Obfuscation\ or\ Information)_{i,t} + Controls_{i,t} + YEAR\ FE + INDUSTRY\ FE + \varepsilon_{i,t}$$

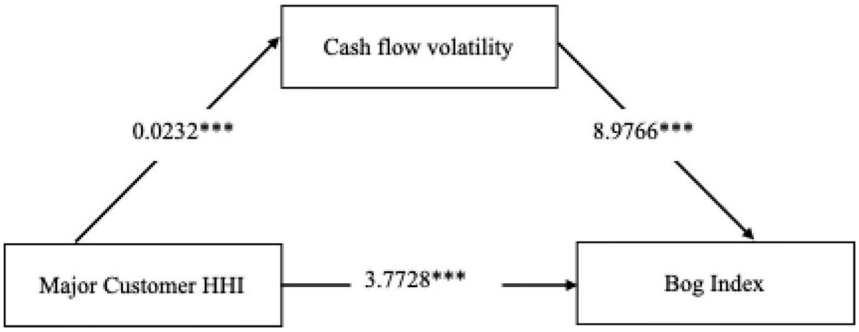
We examine the interaction between customer concentration and the complexity of language used by managers in their disclosures. To measure this complexity, we apply the Gunning Fog index as suggested by Bushee et al. (2018),[8] to the transcripts of conference calls. The obfuscation component, represented by Obfu (Both), combines two proxies: Obfu (Present) and Obfu (Response), which reflect the use of complicated language intended to make disclosures less comprehensible. The information component, labelled Fog (Info (Both)), assesses how much technical and informative content managers provide during earnings calls. The Fog index values are derived from both the presentation part of the call and the responses given during the Q&A session.

Table 9 reports the regression results that indicate that the interactions between customer concentration and the obfuscation component are both positive and statistically significant at the 5 % level for both readability proxies. In contrast, interactions with the

[8] We thank Professor Brian Bushee for sharing the data.

Table 10
Mediation analysis – cash flow volatility.

Bog Index	Path = Cash flow volatility
Direct Path [Major Customer HHI, Cash flow volatility]	3.7728***
	(0.0000)
Mediated Path	
I. [Major Customer HHI, Path]	0.0232***
	(0.0000)
II. [Path, Cash flow volatility]	8.9766***
	(0.0000)



Note: This table reports the path analysis of the relationship among 10-K report readability, cash flow volatility as the mediating variable and customer concentration. The table demonstrates a structural equation model (SEM) of the direct effect of customer concentration on 10-K report readability, as well as the indirect effect through cash flow volatility. The model includes all the baseline control variables. The results remain quantitatively similar without the inclusion of control variables. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

information content are positive but lack statistical significance. This suggests that the obfuscation aspect of linguistic complexity plays a significant moderating role in the positive correlation between customer concentration and the readability of 10-K reports. Therefore, it confirms that firms with a concentrated customer base intentionally lower the readability of their financial disclosures to obscure information rather than providing higher volumes of technical details in their communications.

5.2. Path analysis: cash flow volatility

We further verify whether suppliers with concentrated customers are motivated to reduce 10-k report readability to tackle potential cash flow volatility from losing customers (Dhaliwal et al., 2016; Huang et al., 2016), using a structural equation model (SEM) to explain the direct and indirect effects. Following prior studies, we use path analysis based on SEM to answer the question of how customer concentration as a source variable affects 10-k readability (outcome variable) (Cao et al., 2023). We investigate the direct effect of customer concentration on financial report readability and the indirect effect through cash flow volatility as a mediating variable. We report the results in Table 10, using cash flow volatility as the mediating variable and show the direct and indirect paths through which customer concentration is expected to affect financial report readability.

The direct effect of customer concentration on financial report readability is significantly positive at the 1 % level (3.7728, $p < 0.01$), indicating that customer concentration reduces the readability of financial reports. Additionally, the relationship between customer concentration and cash flow volatility (as proxied by 4 years standard deviation of cash follow ratio measured by cash flow and income taxes paid divided by total assets) is also significantly positive (0.0232, $p < 0.01$), suggesting that customer concentration increases cash flow volatility. Furthermore, the positive, significant relationship between cash flow volatility and financial report readability (8.9766, $p < 0.01$) implies that higher cash flow volatility is associated with reducing the readability of financial reports. Therefore, the path analysis confirms that higher cash flow volatility is a key mechanism through which customer concentration reduces the readability of financial reports.

5.3. Cross-sectional analysis

5.3.1. CEO excess compensation

Managers' relationships and reputation with their key customers are crucial in establishing inherent contracts between suppliers and major customers (Klein & Leffler, 1981). Consistent with this, the connections and reputation of CEOs in customer-reliant suppliers are essential to sustain long-term relationships with significant customers (Johnsen & Lacoste, 2016). Since CEO compensation is often viewed as a result of agency issues, it is likely the primary means of transferring wealth from shareholders to management (Masulis et al., 2009). Moreover, CEOs who receive stock options may face heightened agency problems, particularly when the options are granted unexpectedly, which can be seen as a form of opportunistic compensation that disproportionately benefits the CEO (Hoi, Wu, & Zhang, 2019). For example, if option award dates are set following the announcement of bad news or just before positive news

Table 11
Cross-sectional analysis - CEO Excess Compensation.

Dependent Variable	Bog Index		Ln (File Size)	
	(1)	(2)	(3)	(4)
Major Customer Sales × Excess Compensation	0.7584***		0.0256**	
	(0.000)		(0.041)	
Major Customer HHI × Excess Compensation		1.3958***		0.0489**
		(0.000)		(0.025)
Major Customer Sales	2.1703***		0.1134***	
	(0.000)		(0.000)	
Major Customer HHI		3.5057***		0.1801***
		(0.000)		(0.000)
Excess Compensation	−0.5429***	−0.5401***	−0.0426***	−0.0423***
	(0.000)	(0.000)	(0.000)	(0.000)
Constant	80.8842***	80.9113***	11.9160***	11.9175***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	33,293	33,293	28,040	28,040
Adjusted R-squared	0.4601	0.4618	0.3544	0.3567
Controls	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: This table reports the channel effect of CEO excess compensation on the relationship between customer concentration and financial statement readability. In models 1 and 2, the dependent variable is the Bog Index. Models 3 and 4 consider File Size as the dependent variable. The key variable of interest is the interaction between CEO excess compensation and the customer concentration proxies (major customer sales and HHI). CEO excess compensation is measured as the residuals from the regression of the natural logarithm of CEO total compensation on the natural logarithm of total market value of the firm. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

becomes public, this can enhance the value of the CEO's options (Chauvin & Shenoy, 2001). CEOs acting in their own interest may exploit excessive compensation practices to gain private advantage. We contend that the inverse relationship between the concentration of major customers and the readability of 10-K reports is amplified when CEOs receive excessive compensation.

To quantify CEO excess compensation, we follow prior research and calculate it as the residuals from a regression analysis that examines the natural logarithm of total CEO compensation against the natural logarithm of the firm's total market value firm (Fich et al., 2011; Kim et al., 2023). Our central hypothesis is that the negative relationship previously identified between customer concentration and the readability of financial statements will be intensified when CEOs are overcompensated. We then re-estimate our baseline regressions, incorporating the interaction between CEO excess compensation and the customer concentration measures.

Our results in Table 11 show that the interaction terms between CEO excess compensation and the measures of customer concentration are consistently positive and statistically significant across models 1 to 4. This indicates that the negative relationship between customer concentration and the readability of financial statements becomes more pronounced in scenarios involving managerial agency issues, particularly when CEO compensation is excessive.

5.3.2. Institutional ownership

Institutional investors are widely considered important capital market participants who monitor managerial behaviour and significantly influence financial decisions (Gillan & Starks, 2000; Graham et al., 2005; Koh, 2003; Nguyen, 2021; Shleifer & Vishny, 1986). Rising institutional ownership has been found to improve the quantity, accuracy and informativeness of managerial disclosures (Bird & Karolyi, 2016; Boone & White, 2015; Jain et al., 2023). To improve institutional investor monitoring, improved readability is required because it mitigates information asymmetry (Choi & Chung, 2023). Through greater institutional ownership, managers have less propensity to obfuscate disclosures, which significantly improves the readability of 10-k reports (Chakrabarty et al., 2018). In this conjecture, information asymmetry has a negative implication for supplier firms with a concentrated customer network (Lee et al., 2020). Therefore, we hypothesize that the negative impact of customer concentration on financial report readability will be mitigated in firms with greater institutional monitoring, brought about by institutional ownership. We assess institutional ownership by calculating the percentage of outstanding shares held by institutional investors, using data from the Thomson Reuters Institutional (13F) Holdings database.[9] We then re-evaluate our baseline regression models by incorporating the interaction between institutional ownership and the customer concentration measures.

The results in Table 12 show that the interaction terms between institutional ownership and the customer concentration measures are consistently negative and statistically significant in models 1–4. This indicates that suppliers with a concentrated customer base improve the readability of their financial statements in the presence of institutional owners, exemplifying their monitoring ability of

[9] We thank the anonymous reviewer for this suggestion.

Table 12
Cross-sectional analysis – Institutional Ownership.

Dependent Variable	(1)	(2)	(3)	(4)
	Bog Index		Ln (File Size)	
Major Customer Sales × Institutional Ownership	−1.5350***		−0.0261**	
	(0.000)		(0.039)	
Major Customer HHI × Institutional Ownership		−2.4859***		−0.0449*
		(0.000)		(0.088)
Major Customer Sales	2.2294***		0.1147***	
	(0.000)		(0.000)	
Major Customer HHI		3.3938***		0.1657***
		(0.000)		(0.000)
Institutional Ownership	0.6854***	0.4986***	0.0091*	0.0064*
	(0.000)	(0.000)	(0.078)	(0.064)
Constant	72.2758***	79.0296***	11.9686***	11.9703***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	75,660	75,660	58,893	58,893
Adjusted R-squared	0.4234	0.4230	0.4194	0.4192
Controls	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: This table reports the channel effect of institutional ownership on the relationship between customer concentration and financial statement readability. In models 1 and 2, the dependent variable is the Bog Index. Models 3 and 4 consider File Size as the dependent variable. The key variable of interest is the interaction between institutional ownership and the customer concentration proxies (major customer sales and HHI). Institutional ownership is measured as the percentage of firm stocks owned by institutional owners. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

opportunistic managerial behaviour.

5.3.3. Customer switching costs

One of the biggest risk factors suppliers face in having a focussed customer base is major customers easily switching suppliers (low switching cost), leading to material financial losses and increased probability of facing financial distress (Dhaliwal et al., 2016; Huang et al., 2016). When suppliers rely on major customers who can easily switch to alternative suppliers, they may encounter heightened uncertainty, leading to fluctuating cash flows and potential financial distress. Research indicates that disruptions in the relationships between customers and suppliers can result in negative abnormal stock returns, as well as declines in financial performance, resilience and sales growth (Chen et al., 2021; Intintoli et al., 2017; Jiang et al., 2023). Suppliers with a concentrated customer base face significantly higher proprietary disclosure costs when it comes to ending a customer-supplier relationship. Consequently, dependent suppliers may opt to postpone disclosing bad news to maintain existing supply chain connections. We anticipate that such suppliers will produce less readable financial reports with the intent to obfuscate (Cen et al., 2017; Ertugrul et al., 2017; Lo et al., 2017).

In line with previous research by Dhaliwal et al. (2016) and Huang et al. (2016), we establish two metrics to gauge supplier sales-based industry market share and assess customer switching costs. The first metric captures the percentage of total sales in the supplier's 2-digit SIC industry that the supplier commands. However, because suppliers in the same 2-digit SIC category may not compete for identical customers, we calculate a second metric reflecting the percentage of the supplier's overall 3-digit SIC industry revenue that they actually obtain. To identify instances where customers face low switching costs, we define a dummy variable that equals one if the metric is below the annual sample median and zero otherwise. We then re-evaluate our baseline regression models by including the interaction between the low switching cost variables and the customer concentration measures.

Table 13 reports the results on the influence of customer switching costs on the link between customer concentration and the readability of 10-K reports. Panel A focuses on findings related to the 2-digit SIC switching cost variable and Panel B displays the results based on the 3-digit SIC code. Our analysis reveals that the interaction terms between low switching costs and customer concentration measures are consistently positive and statistically significant across both panels. Furthermore, the low switching cost variable shows individual positive and significant coefficients for both the BOG Index and file size in Panels A and B. These results suggest that the negative effect of customer concentration on the quality of financial statements is amplified when major customers can easily switch suppliers at minimal costs.

5.3.4. Board independence

Studies report conflicting evidence regarding how board independence influences the readability of annual reports. It is often argued that an independent board promotes effective monitoring, leading to more disclosures, an improved information environment and better transparency (Adams & Ferreira, 2007; Armstrong et al., 2010; Drymiotes, 2011; Goh et al., 2016). Conversely, monitoring through board independence may be avoided by opportunistic managers through information obfuscation since studies have found

Table 13
Cross-sectional analysis - Customer switching cost.

Panel A: Supplier Market Share Based on 2-digit SIC Industries				
Dependent Variable	(1)	(2)	(3)	(4)
	Bog Index		File Size	
Major Customer Sales	1.6294***		0.1130***	
	(0.000)		(0.000)	
Major Customer Sales × Low Switching Costs	1.5574***		0.0516***	
	(0.000)		(0.009)	
Major Customer HHI		2.4050***		0.1451***
		(0.000)		(0.000)
Major Customer HHI × Low Switching Costs		1.9188***		0.0893***
		(0.000)		(0.005)
Low Switching Cost	0.8090***	0.8556***	0.0056**	0.0064**
	(0.000)	(0.000)	(0.028)	(0.021)
Constant	78.1425***	78.1312***	11.8947***	11.8963***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	107,248	107,248	79,872	79,872
Adjusted R-squared	0.3985	0.3848	0.4475	0.4197
Control Variables	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Panel B: Supplier Market Share Based on 3-digit SIC Industries				
Dependent Variable	(1)	(2)	(3)	(4)
	Bog Index		File Size	
Major Customer Sales	1.6694***		0.1269***	
	(0.000)		(0.000)	
Major Customer Sales × Low Switching Costs	1.2572***		0.0167**	
	(0.000)		(0.044)	
Major Customer HHI		2.4937***		0.1746***
		(0.000)		(0.000)
Major Customer HHI × Low Switching Costs		1.4322***		0.0238**
		(0.000)		(0.046)
Low Switching Cost	1.3659***	1.4106***	0.0427***	0.0434***
	(0.000)	(0.000)	(0.000)	(0.000)
Constant	77.5997***	77.5877***	11.8549***	11.8564***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	107,248	107,248	79,872	79,872
Adjusted R-squared	0.4009	0.4004	0.4481	0.4478
Control Variables	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: This table presents the results of the impact of customer concentration on financial statement readability based on customer switching cost. Panel A presents the findings based on Supplier Market Share Based on 2-digit SIC Industries samples and Panel B presents the findings based on Supplier Market Share Based on 3-digit SIC Industries. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

that greater independent director presence leads to reduced annual report readability (Rahman & Kabir, 2023). However, from a customer-supplier relationship perspective, a greater proportion of independent directors is found to restrict firms from acquiring value destroying customer-concentrated firms (Dong et al., 2021). Therefore, we expect some relationship to be played by board independence in the linkage between customer concentration and 10-k report readability, though the direction is unclear. We measure board independence as the ratio of independent directors to the total number of board members, with data from BoardEx[10]. We re-estimate our baseline regressions after including the interaction between board independence with the customer concentration proxies.

Table 14 shows that the interactions in models 1 to 4 have positive, statistically significant relationships with the readability proxies as dependent variables. This implies that the existing negative effect of customer concentration and 10-k report readability is

[10] We thank the anonymous reviewer for this suggestion.

Table 14
Cross-sectional analysis – Board Independence.

Dependent Variable	(1)	(2)	(3)	(4)
	Bog Index		Ln (File Size)	
Major Customer Sales × Board Independence	4.6983***		0.0560**	
	(0.000)		(0.047)	
Major Customer HHI × Board Independence		5.9250***		0.1343*
		(0.000)		(0.072)
Major Customer Sales	1.4312*		0.0316	
	(0.064)		(0.551)	
Major Customer HHI		1.4967		0.0056
		(0.221)		(0.947)
Board Independence	0.2894	0.4188**	0.0038	0.0039
	(0.154)	(0.034)	(0.796)	(0.786)
Constant	81.8002***	81.7588***	12.1311***	12.1332***
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	43,362	43,362	37,494	37,494
Adjusted R-squared	0.3947	0.3944	0.3621	0.3637
Controls	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES

Note: This table reports the channel effect of Board Independence on the relationship between customer concentration and financial statement readability. In models 1 and 2, the dependent variable is the Bog Index. Models 3 and 4 consider File Size as the dependent variable. The key variable of interest is the interaction between board independence and the customer concentration proxies (major customer sales and HHI). Board independence is measured as the ratio of independent directors in the board to total number of board members. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

more pronounced for suppliers with greater board independence. This could be through the greater managerial obfuscation incentives while dealing with independent boards, in line with prior studies (Rahman & Kabir, 2023).

5.4. Subsample analysis: relation-specific investments

Suppliers with a concentrated customer base often undertake sizable investment to particularly cater to their major customer's specific needs that might not have much value outside the existing customer-supplier relationship. Such investment might lead to substantial losses if a supplier loses its major customer, exposing the supplier to the considerable probability of financial distress. This probability might become more pronounced if the major customer implements strict bargaining power with the supplier (Campello & Gao, 2017; Dong et al., 2021; Kale & Shahrur, 2007; Titman & Wessels, 1988). Supplier firms with higher levels of asset specificity would be more likely to produce less readable annual reports when there are possible disruptions to the existing customer-supplier relationship (Sun, 2023; Weisskopf, 2024).

Following prior studies (Crawford et al., 2020; Hui et al., 2012; Kale & Shahrur, 2007; Raman & Shahrur, 2008), we perform subsample analysis using two indicators of relationship-specific investments: R&D intensity, which is calculated as the ratio of R&D expenditure to total assets for firms with available R&D data in Compustat; and investment in unique assets, defined as the sum of selling, general and administrative (SG&A) expenses and advertising costs relative to total assets. Our main premise is that the previously observed negative correlation between customer concentration and the readability of financial statements will be amplified when suppliers make substantial relationship-specific investments. As the supplier's relationship with a major customer becomes more focused on that customer's distinct product needs, the customer is likely to demand more information to assess their claims against the supplier. Consequently, suppliers may feel compelled to provide more information directly to the customer, which could lead to less information being publicly disclosed, thereby diminishing the quality of their financial statements. To investigate this relationship, we re-estimate our baseline regression models for the Major Customer Sales and Major Customer HHI concentration metrics and compare them against the readability proxies by creating subsamples of firm-year observations for suppliers with relationship-specific investments above and below the median.

Table 15 presents the results of the cross-sectional analysis looking at the effects of the relationship-specific investments; Panel A presents the results for R&D intensity and Panel B presents the results of the selling expenses-based subsamples. Panel A shows significant positive coefficients for Major Customer Sales and Major Customer HHI on the readability proxies for the subsample of firms with higher than the median relationship-specific investment proxies. However, for the lower than median subsamples of relation-specific proxies, the coefficients of the customer concentration measures are much lower in value, and statistically less significant across all models. F-tests conducted for all the subsamples determine that the differences between their coefficients are significant in all models. Overall, these results corroborate our baseline findings and support the robustness of our results that the negative impact of customer concentration on financial statement readability is more pronounced when suppliers commit to higher relation-specific

Table 15
Subsample analysis – relationship-specific investment.

Panel- A: R&D Intensity

Dependent Variable	Bog Index		File Size		Bog Index		File Size	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	High R&D	Low R&D	High R&D	Low R&D	High R&D	Low R&D	High R&D	Low R&D
Major Customer Sales	2.8275***	0.9247*	0.1533***	0.0836*				
	(0.000)	(0.061)	(0.000)	(0.055)				
Major Customer HHI					3.9171***	1.4456	0.2107***	0.1210*
					(0.000)	(0.307)	(0.004)	(0.066)
Constant	82.4456***	77.0586***	12.0498***	11.7947***	82.4984***	77.0612***	12.0537***	11.7953***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Control Variables	YES	YES	YES	YES	YES	YES	YES	YES
Observations	46,550	60,696	35,045	44,824	46,550	60,696	35,045	44,824
Adjusted R-squared	0.3589	0.4042	0.3810	0.4766	0.3580	0.4041	0.4192	0.4766
P-value for the F-test for:	0.012**		0.031**					
$\beta_{left}>\beta_{right}$					0.008***		0.062*	
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES	YES

Panel – B: Selling Expenses

Dependent Variable	Bog Index		File Size		Bog Index		File Size	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	High Selling Expenses	Low Selling Expenses	High Selling Expenses	Low Selling Expenses	High Selling Expenses	Low Selling Expenses	High Selling Expenses	Low Selling Expenses
Major Customer Sales	2.7305***	0.9089*	0.1536***	0.1076*				
	(0.000)	(0.084)	(0.000)	(0.053)				
Major Customer HHI					3.9569***	2.6097*	0.2215***	0.1407**
					(0.000)	(0.078)	(0.000)	(0.032)
Constant	79.3000***	78.3740***	11.9257***	11.7658***	79.3213***	78.4036***	11.9274***	11.7683***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Control Variables	YES	YES	YES	YES	YES	YES	YES	YES
Observations	68,262	38,983	49,468	30,403	68,262	38,983	49,468	30,403
Adjusted R-squared	0.4070	0.4013	0.4452	0.4544	0.4066	0.4009	0.4450	0.4541
P-value for the F-test for:	0.028**		0.041**					
$\beta_{left}>\beta_{right}$					0.015**		0.071*	
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES	YES

Note: This table presents the results of impact of customer concentration on financial statement readability based on relationship-specific investment. Panel A presents the findings based on R&D Intensity (R&D Expenditure scaled by book assets) as the relationship-specific investment indicator and Panel B presents the findings based on Selling Expenses (sum of supplier SG&A expenses and advertising expenses scaled by book assets) as the relationship-specific investment indicator. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

investments.

5.5. The outcome of high customer concentration and lower readability – firm value and stock price crash risk

Following our analysis that shows a robust, consistently negative association between customer concentration and 10-k report readability, we explore some consequences of the phenomenon on firm-level performance outcomes.[11] We first look at the outcome of higher customer concentration and poor financial statement readability on firm value, proxied by Tobin's Q. Studies have shown that poor readability motivated by the managerial obfuscation motive is more common in bad performing firms (Hasan, 2020; F. Li, 2008).

[11] We thank the anonymous reviewer for this suggestion.

Table 16
Outcome of high customer concentration and lower quality financial reports on firm value (Tobin's Q) and stock price cash risk (NCSKEW and DUVOL).

Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
	Tobin's Q		$NCSKEW_{t+1}$		$DUVOL_{t+1}$	
Major Customer Sales * Above Median Bog Index	−0.3409**		0.0349**		0.0207**	
	(0.043)		(0.047)		(0.034)	
Major Customer Sales * Above Median File Size		−0.3284**		0.0099*		0.0006*
		(0.045)		(0.057)		(0.095)
Above Median Bog Index	0.0709		0.0086***		0.0046***	
	(0.289)		(0.007)		(0.009)	
Above Median File Size		0.2289		0.0033		0.0034**
		(0.251)		(0.287)		(0.048)
Major Customer Sales	0.1872	0.1907	−0.0143	−0.0008	−0.0178	−0.0060
	(0.165)	(0.185)	(0.303)	(0.995)	(0.321)	(0.443)
Constant	3.1932***	3.1571***	0.3187***	0.3177***	0.1951***	0.1911***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Observations	107,277	107,277	75,409	75,409	75,381	75,381
Adjusted R-squared	0.3463	0.3467	0.0520	0.0650	0.0581	0.0582
Controls	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES

Note: This table presents the results examining the outcome of high customer concentration and lower quality financial reports on firm value (proxied by Tobin's Q) and stock price crash risk (proxied by NCSKEW and DUBOL) Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

Equity markets also react more positively to more readable reports, driven by a reduction in future performance related uncertainty (Hsieh et al., 2016). Customer concentration has a negative impact on a supplier's profitability at the expense of better customer performance and the supplier's own CSR performance (Hui et al., 2019; Zhu et al., 2021). Therefore, we expect suppliers with a concentrated customer base and lower readability to demonstrate poor financial performance and firm value.

Moreover, consistent with the notion of managers hiding bad news through complex financial reports, less readable 10-k reports are associated with higher stock price crash risk (Ertugrul et al., 2017; Kim et al., 2019). In addition, whereas suppliers with a concentrated government customer base reduce stock price crash risk, having a concentrated corporate customer network has a significantly higher crash risk. This effect is more pronounced for suppliers with higher relationship-specific investments and lower customer switching costs (Lee et al., 2020; Ma et al., 2020). We have already shown how higher customer concentration with greater relationship-specific investments and lower switching costs leads to reduced readability. Therefore, we also expect suppliers to face greater stock price crash risk when they have an undiversified corporate customer base and poor readability of annual reports.

We test these hypothesized outcomes and report the results in Table 16. The dependent variables are proxies for firm value (Tobin's Q) and stock price crash risk (NCSKEW and DUVOL), measured following prior studies (Hasan, 2020; Ma et al., 2020). Following Ma et al. (2020), we take one period lead values of the crash risk proxies. To consider poor readability, we construct dummy variables taking the value of one if the readability proxies (Bog index and File Size) have values above their medians. The variable of interest is the interaction between the customer concentration and poor readability proxies. The results in Table 16 show that the interactions in models 1 and 2 with Tobin's Q as the dependent variable are negative and statistically significant at the 5 % level. This implies that suppliers with high customer concentration and poor readability of their 10-K reports demonstrate poor financial performance. The results from models 3–6 show that the interaction variable is positive and statistically significant. Based on these coefficients, we can determine that suppliers with higher customer concentration and poor 10-k report readability face greater stock price crash risk. These results further validate the study's findings and clarify how greater customer concentration and poor financial reporting quality are detrimental to firm performance.

6. Conclusion

We have investigated the association between a supplier's customer concentration and financial report quality. Using a large sample of US firms, we find consistent, robust validation that suppliers with a highly concentrated customer base produce less readable 10-K reports. We conduct 2SLS regressions using the instrumental variable approach to validate that our findings are robust to endogeneity concerns. Using propensity score matching and entropy balanced samples as well as the Oster (2019) test, we address concerns regarding omitted variable bias and model misspecification issues. We further identify the managerial obfuscation motive and cash flow volatility to drive this association, in line with prior literature (Bushee et al., 2018; Dhaliwal et al., 2016).

We find corroborating evidence that demonstrates that the negative association between customer concentration and the readability of financial reports is stronger when the supplier firm's CEO receives excess compensation, faces greater board independence

and lower switching costs and when the supplier commits to greater relationship-specific investment. However, we find institutional monitoring to significantly improve the readability of 10-K reports for suppliers with concentrated customers, providing a possible tool for stakeholders to hold managers to accountable in such circumstances. These findings contribute to the literature on the readability of corporate disclosures and strengthen customer-supplier network studies in accounting and finance.

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Appendices

Appendix Table 1
Variable definitions

Variable	Formula/derivation	Data Source
Dependent Variables		
BOG Index	BOG Index is a comprehensive measure of a document's plain English problems, including passive voice, redundant verbs, use of jargon, and sentence complexity, among others (Bonsall & Miller, 2017). A higher Bog Index reflects worse document readability.	Author personal website
File Size	Net file size (in megabytes) of the 10-K document (Loughran & McDonald, 2014)	Author personal website
Fog index	Fog Index measures readability by adding the average sentence length and the percentage of long words present in the text (Loughran & McDonald, 2014)	Author personal website
Independent Variables		
Major Customer Sales	Measured as the fraction of a supplier's annual total sales obtained by all customers who account for at least 10 % of the supplier's annual revenues (Banerjee et al., 2008; Dhaliwal et al., 2016).	Compustat customer-segment file
Major Customer HHI	Calculates supplier *i*'s customer concentration in year *t* across the supplier's *J* major clients as follows: $Major\ Customer\ HHI_{it} = \sum_{j=1}^{J}\left(\frac{Sales_{ijt}}{Sales_{it}}\right)^2$ where: $Sales_{it}$ denotes a supplier's total sales in year *t* and $Sales_{ijt}$ denotes a supplier's sales to a large client *j* in year *t*.	Compustat customer-segment file
Major Customer	A dummy variable equal to 1 if the supplier firm has an identifiable major customer representing at least 10 % of the supplier's sales and 0 otherwise.	Compustat customer-segment file
Control Variables		
Leverage	Long-term (total) debt plus current (total) liabilities to the total book value of assets	Compustat
Book to Market Value	Book value of equity to market value of equity	Compustat
Size	Natural logarithm of total assets	Compustat
Return on Asset (ROA)	Operating income before depreciation to the total book value of assets	Compustat
Intangible Assets	Intangible assets to total book value of assets	Compustat
Asset Tangibility	Tangible assets (net fixed assets) to total assets	Compustat
Cash Flow to Assets	Cash and short-term investments scaled by total book value of assets	Compustat
Geographic Segments	Natural log of the number of geographic segments	Compustat customer-segment file
Business Segments	Natural log of the number of business segments	Compustat customer-segment file
Firm Age	Natural log of the age of the firm	Compustat and author personal website (Appels & Kowalzick, 2021)
Free Cash Flow	Operating income before depreciation minus interest expenses minus income taxes minus capital expenditure scaled by the book value of total assets	Compustat
Mean Analyst Following	Arithmetic mean of unique monthly earnings forecasts during each calendar year for each firm	I/B/E/S
Variables used in additional analyses		
CEO excess compensation	CEO excess compensation is measured as the residuals from the regression of the natural logarithm of CEO total compensation on the natural logarithm of total market value of the firm	Compustat
R&D Intensity	R&D expenditures to total assets for firms with non-missing R&D expenditures in Compustat	Compustat
Selling Expenses	SG&A and advertising expenses to total assets	Compustat
BadCustIndShock	BadCustIndShock is defined as the weighted average of an indicator variable, which equals one for major customers experiencing an annual stock return drop of 10 % or more, reflecting an adverse industry shock, and zero otherwise, adjusted by the percentage of sales to each corresponding major customer.	CRSP
Obfuscation (Both)	Obfuscation (Both) represents the estimated latent obfuscation component derived from the first principal component of Obfu (Present), which pertains to the presentation portion of the call, and Obfu (Response), which refers to the response portion of the call, following the methodology outlined by Bushee et al. (2018).	Bushee et al. (2018)

(*continued on next page*)

Appendix Table 1 (*continued*)

Variable	Formula/derivation	Data Source
Information (Both)	The estimated latent information component, based on Bushee et al. (2018), encompasses both Info (Present), which pertains to the presentation portion of the earnings call, and Info (Response), which relates to the response portion, with Information (Both) representing the first principal component derived from these two measures.	Bushee et al. (2018)
Cash flow volatility	4 years standard deviation of cash follow ratio measured as (cash flow + income taxes paid)/ total assets	
Institutional Ownership	Percentage of shares outstanding held by institutional investors	Thomson Reuters
Board Independence	Number of independent directors scaled by total number of board members	BoardEx
Low Switching Costs	A dummy variable for the low customer switching cost set equal to 1 if the measure is below the sample median for that year and 0 otherwise	Compustat
Tobin's Q	Tobin's Q ratio is measured by market value of assets scaled by replacement cost of capital	Compustat
NCSKEW	NCSKEW is the negative skewness of weekly stock returns calculated following prior studies (Hasan, 2020; Ma et al., 2020)	CRSP
DUVOL	DUVOL is the ratio of upward and downward fluctuations in stock returns calculated following prior studies (Hasan, 2020; Ma et al., 2020)	CRSP
Board Size	Logarithm of total number of board directors	BoardEx
Board Gender Diversity	Number of female directors scaled by total number of directors	BoardEx

Appendix Table 2
Correlation matrix

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Bog Index	1.00														
2. File Size	0.45	1.00													
3. Major Customer Sales	0.06	0.03	1.00												
4. Major Customer HHI	0.07	0.03	0.84	1.00											
5. Leverage	−0.01	−0.02	−0.01	−0.01	1.00										
6. Book to Market Ratio	−0.01	−0.01	−0.01	−0.04	−0.02	1.00									
7. Size	0.18	0.42	0.01	−0.02	−0.06	−0.01	1.00								
8. ROA	−0.11	0.03	0.05	0.03	−0.03	0.01	0.43	1.00							
9. Intangible Assets	0.11	0.12	−0.02	−0.03	−0.02	−0.03	0.07	0.09	1.00						
10. Asset Tangibility	−0.16	0.06	0.01	0.01	−0.02	−0.01	0.22	0.20	−0.24	1.00					
11. Cash Flow to Assets	0.27	0.02	0.10	0.11	−0.02	−0.02	−0.20	−0.18	−0.18	−0.34	1.00				
12. Geographic Segments	0.13	0.20	0.05	0.01	−0.03	−0.04	0.33	0.16	0.10	−0.06	−0.01	1.00			
13. Business Segments	0.13	0.24	0.06	0.02	−0.02	−0.03	0.35	0.15	0.08	−0.04	−0.02	0.64	1.00		
14. Ln (Firm Age)	0.05	0.23	−0.03	−0.04	−0.01	−0.04	0.40	0.22	0.03	0.09	−0.17	0.32	0.35	1.00	
15. Free Cash Flow	−0.05	0.02	0.04	0.03	−0.05	0.02	0.37	0.56	0.10	0.04	−0.09	0.17	0.16	0.18	1.00

Appendix Table 3
Baseline regression results with additional governance control variables

Dependent Variable	Bog Index			Ln (File Size)		
	(1)	(2)	(3)	(4)	(5)	(6)
Major Customer Sales	1.6747***			0.0705***		
	(0.000)			(0.000)		
Major Customer HHI		2.4212***			0.0970***	
		(0.000)			(0.000)	
Major Customer			0.7011***			0.0381***
			(0.000)			(0.000)
Leverage	1.6647***	1.6617***	1.6632***	0.1604***	0.1603***	0.1602***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Book to Market Ratio	0.0004	0.0004	0.0005	0.0001	0.0001	0.0001
	(0.660)	(0.666)	(0.661)	(0.833)	(0.836)	(0.830)
Size	0.8268***	0.8261***	0.8291***	0.1094***	0.1093***	0.1095***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ROA	−8.5254***	−8.5339***	−8.5042***	−0.3229***	−0.3230***	−0.3231***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Intangible Assets	2.6632***	2.6697***	2.6266***	0.1644***	0.1646***	0.1620***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Asset Tangibility	0.0305	0.0281	0.0264	0.0195	0.0194	0.0189
	(0.874)	(0.884)	(0.891)	(0.157)	(0.159)	(0.170)
Cash Flow to Assets	4.5831***	4.5805***	4.6076***	0.1731***	0.1732***	0.1734***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Geographic Segments	0.0511***	0.0513***	0.0486***	0.0014*	0.0014*	0.0012
	(0.000)	(0.000)	(0.000)	(0.093)	(0.092)	(0.127)
Business Segments	−0.0756***	−0.0745***	−0.0774***	0.0027***	0.0027***	0.0026***

(*continued on next page*)

Appendix Table 3 (*continued*)

Dependent Variable	Bog Index			Ln (File Size)		
	(1)	(2)	(3)	(4)	(5)	(6)
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Ln (Firm Age)	−0.5520***	−0.5522***	−0.5550***	−0.0502***	−0.0500***	−0.0503***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Free Cash Flow	5.7221***	5.7463***	5.6752***	−0.0457	−0.0446	−0.0484
	(0.000)	(0.000)	(0.000)	(0.142)	(0.151)	(0.119)
Board Size	−0.5608***	−0.5670***	−0.5626***	−0.0724***	−0.0721***	−0.0727***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Board Gender Diversity	1.9053***	1.9191***	1.9000***	0.0111	0.0117	0.0109
	(0.000)	(0.000)	(0.000)	(0.591)	(0.573)	(0.601)
Board Independence	0.7873***	0.7963***	0.7798***	−0.0088	−0.0083	−0.0095
	(0.000)	(0.000)	(0.000)	(0.524)	(0.545)	(0.489)
Constant	80.4037***	80.4301***	80.3838***	12.0123	12.0140***	12.0091***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Observations	43,362	43,362	43,362	37,494	37,494	37,494
Adjusted R-squared	0.3954	0.3952	0.3956	0.3634	0.3630	0.3637
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES

Note: This table re-estimates the baseline regressions for customer concentration and financial statement readability, with the inclusion of board size, board gender diversity and board independence as additional governance-related control variables. In models 1 to 3, the dependent variable is the *Bog Index*. Models 4 to 6 are developed by considering *Ln* (*File Size*) as the dependent variable. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

Appendix Table 4
Fog index as the alternative proxy for readability

Dependent Variable	Fog Index		
	(1)	(2)	(3)
Major Customer Sales	0.0414**		
	(0.039)		
Major Customer HHI		0.0752**	
		(0.047)	
Major Customer			0.0157*
			(0.094)
Constant	19.0780***	19.0781***	19.0780***
	(0.000)	(0.000)	(0.000)
Observations	107,277	107,277	107,277
Adjusted R-squared	0.5195	0.5201	0.5198
Controls	YES	YES	YES
Year FE	YES	YES	YES
Industry FE	YES	YES	YES

Note: This table re-estimates the baseline regressions for customer concentration and financial statement readability, with the use of Fog index as an alternative proxy for readability. In models 1 to 3, the dependent variable is the Fog Index. Year and industry fixed effects are included in all models. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

Appendix Table 5
2*SLS* (IV – Lagged industry customer concentration)

Panel-A: First Stage Regressions			
Dependent Variable	Industry Concentration Measures Used as Instrumental Variables		
	Industry Major Customer Sales	Industry Major Customer HHI	Industry Major Customer
	(1)	(2)	(3)
2-year lagged Industry Concentration Measure	0.6272***	0.5858***	0.5879***
	(0.000)	(0.000)	(0.000)
3-year lagged Industry Concentration Measure	0.1504***	0.1403***	0.1539***
	(0.000)	(0.000)	(0.009)
Leverage	−0.0001	−0.0001	0.0002

(*continued on next page*)

Appendix Table 5 (*continued*)

Panel-A: First Stage Regressions

Dependent Variable	Industry Concentration Measures Used as Instrumental Variables		
	Industry Major Customer Sales	Industry Major Customer HHI	Industry Major Customer
	(1)	(2)	(3)
	(0.849)	(0.795)	(0.939)
Book to Market Ratio	0.0002	0.0001	−0.0001
	(0.866)	(0.685)	(0.470)
Size	0.0031***	0.0016***	0.0099***
	(0.000)	(0.000)	
ROA	0.0393***	0.0244***	0.1049***
	(0.000)	(0.000)	
Intangible Assets	−0.0005	−0.0071***	0.0586***
	(0.874)	(0.001)	
Asset Tangibility	0.0117***	0.0055***	0.0450***
	(0.000)	(0.004)	
Cash Flow to Assets	0.0016	−0.0002	−0.0071
	(0.640)	(0.944)	(0.408)
Geographic Segments	−0.0004	−0.0002	0.0014***
	(0.775)	(0.102)	(0.001)
Business Segments	−0.0004	−0.0002*	0.0007*
	(0.784)	(0.073)	(0.057)
Ln (Firm Age)	−0.0069***	−0.0044***	−0.0130***
	(0.000)	(0.000)	(0.000)
Free Cash Flow	−0.0003	−0.0015	0.0074
	(0.939)	(0.511)	(0.421)
Constant	−0.0061	0.0024	−0.0852
	(0.923)	(0.952)	(0.600)
Observations	107,277	107,277	107,277
Adjusted R-squared	0.1137	0.0781	0.1619
Year FE	YES	YES	YES
Industry FE	YES	YES	YES
Underidentification Test (p-value)	2766.566 (0.000)	2384.339 (0.000)	1762.666 (0.000)
Cragg-Donald Wald F Statistic	1432.448	1228.263	900.566

Panel-B: Second Stage Regressions – Dependent Variables: ***Bog Index*** and ***File Size***

Dependent Variable	Bog Index			Ln (File Size)		
	(1)	(2)	(3)	(4)	(5)	(6)
Predicted Major Customer Sales	8.5592***			0.4217***		
	(0.000)			(0.000)		
Predicted Major Customer HHI		27.8501***			1.2832***	
		(0.000)			(0.000)	
Predicted Major Customer			0.6890***			0.0475***
			(0.000)			(0.000)
Leverage	0.0001	0.0002	−0.0005	0.0005*	0.0006**	0.0004*
	(0.792)	(0.625)	(0.985)	(0.054)	(0.040)	(0.085)
Book to Market Ratio	−0.0004	−0.0004	−0.0003	−0.0004	−0.0006	−0.0002
	(0.211)	(0.179)	(0.237)	(0.849)	(0.800)	(0.910)
Size	1.1427***	1.1435***	1.1372***	0.1459***	0.1460***	0.1456***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ROA	−4.2266***	−4.6183***	−3.8718***	−0.2972***	−0.3126***	−0.2843***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Intangible Assets	3.2774***	3.5252***	3.1825***	0.2610***	0.2743***	0.2537***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Asset Tangibility	−4.5660***	−4.5929***	−4.5029***	−0.0274***	−0.0269***	−0.0263***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.002)	(0.002)
Cash Flow to Assets	−4.3473***	−4.1874***	−4.4177***	−0.1749***	−0.1671***	−0.1778***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Geographic Segments	0.1247***	0.1271***	0.1284***	−0.0017***	−0.0015***	−0.0016***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.001)
Business Segments	−0.1221***	−0.1240***	−0.1154***	0.0035***	0.0035***	0.0036***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Ln (Firm Age)	−0.2466***	−0.1687***	−0.3073***	−0.0376***	−0.0338***	−0.0405***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Free Cash Flow	1.6919***	1.7846***	1.6122***	−0.1215***	−0.1183***	−0.1234***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Constant	78.8875***	78.4267***	79.3359***	11.8988***	11.8753***	11.9230***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

(*continued on next page*)

Appendix Table 5 (*continued*)

Panel-B: Second Stage Regressions – Dependent Variables: ***Bog Index*** and ***File Size***						
Dependent Variable	Bog Index			Ln (File Size)		
	(1)	(2)	(3)	(4)	(5)	(6)
Observations	107,277	107,277	124,328	93,097	93,097	93,097
R-squared	0.1410	0.0678	0.0685	0.2967	0.3236	0.3606
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES

Note: The findings of a two-stage lease-squares regression analysis using the instrumental variable of Lagged Industry Concentration Measures are shown in this table. The findings of the first stage OLS regression analysis are presented in Panel A and second stage models are presented in Panel B. Standard errors are heteroscedasticity-adjusted and clustered at the firm and year levels and P-values are shown in parenthesis. Significance at the 10%, 5% and 1% level is indicated by *, **, and ***, respectively.

Data availability

Data will be made available on request.

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